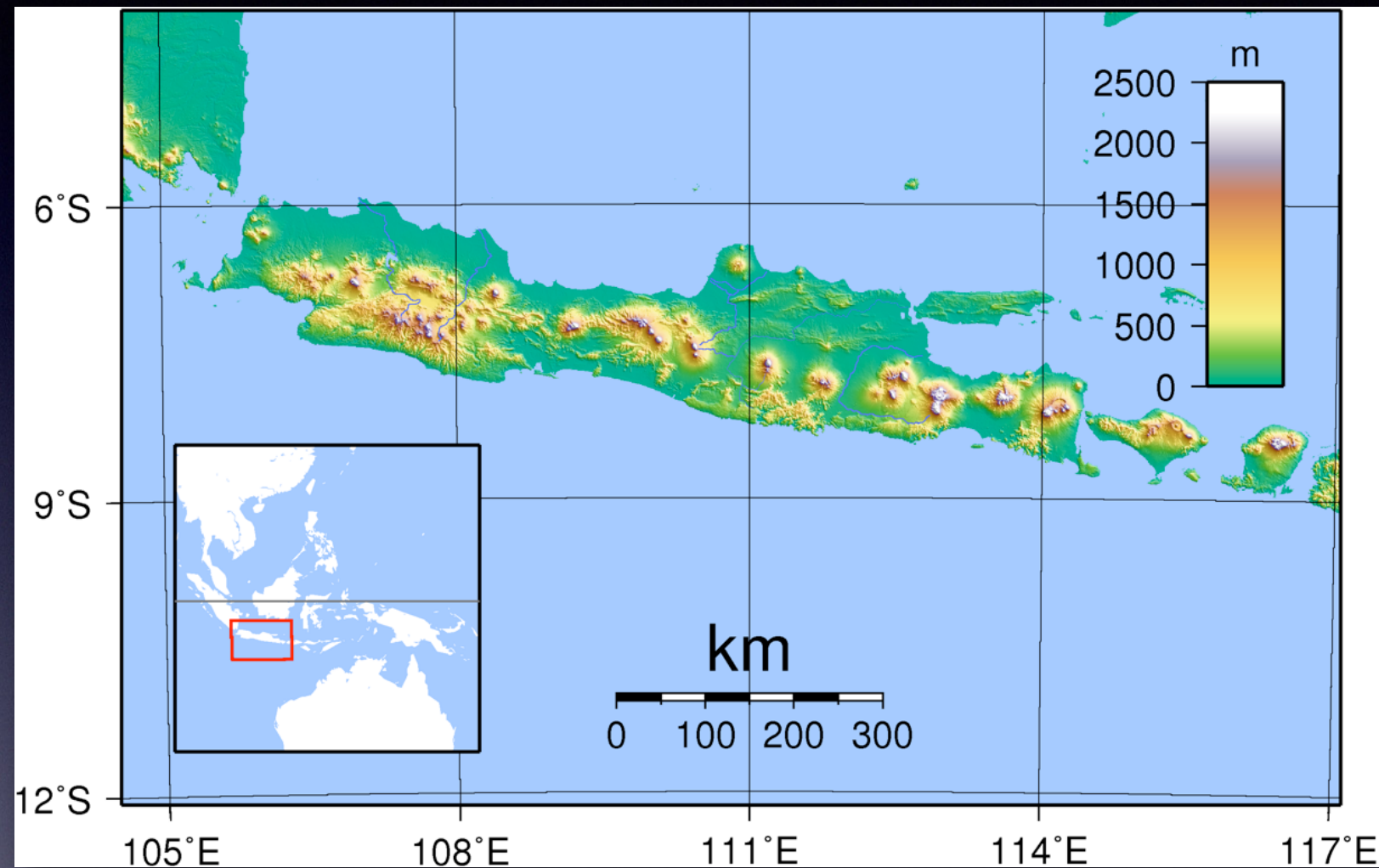


JAVA



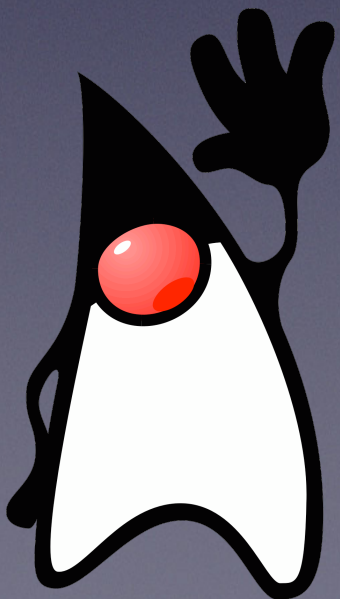
Con una población de 135 millones de personas, la isla de Java, Indonesia, es la isla más poblada del mundo.



El lenguaje de programación



El lenguaje de programación

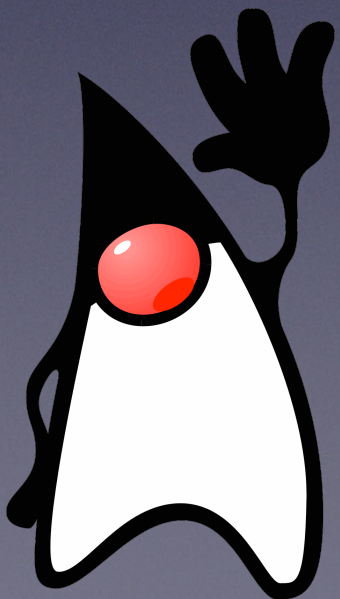


Duke es la mascota de Java



oo

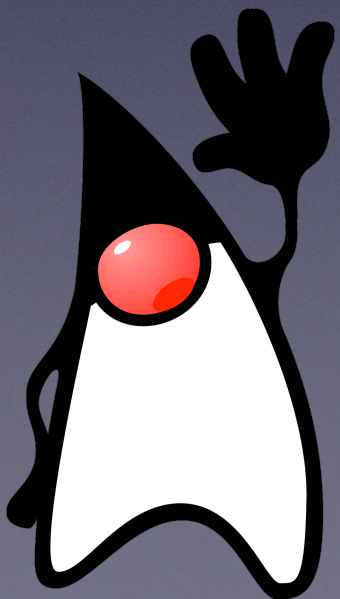
Lenguaje Orientado a Objetos





OO

Sin punteros (manejados por el programador)

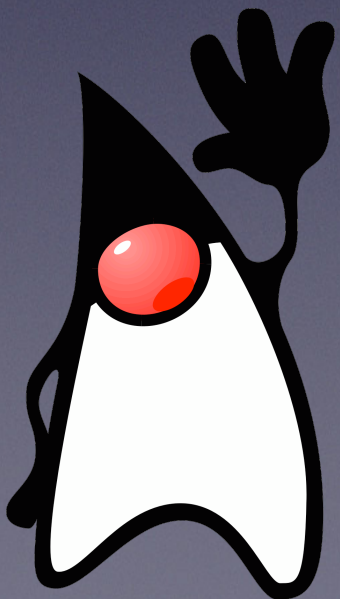




OO

Sin punteros (manejados por el programador)

Manejo sencillo de memoria (recolector de basura)



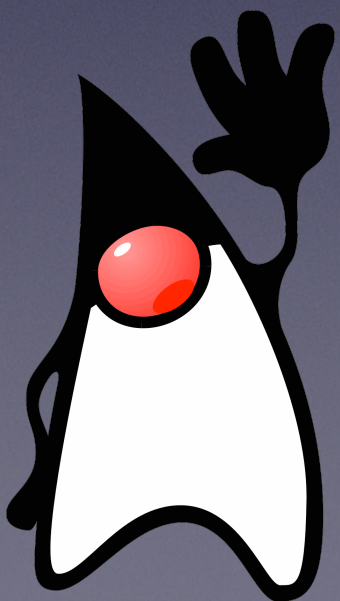


OO

Sin punteros (manejados por el programador)

Manejo sencillo de memoria (recolector de basura)

Genera programas con pocos errores





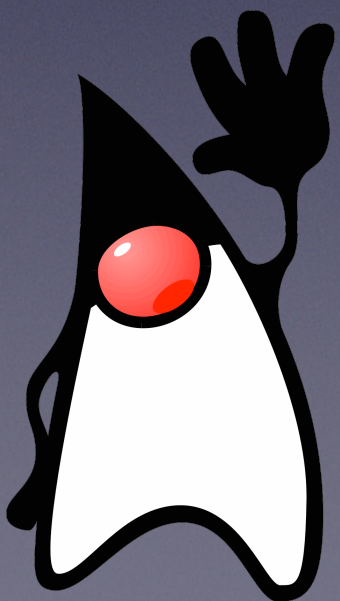
OO

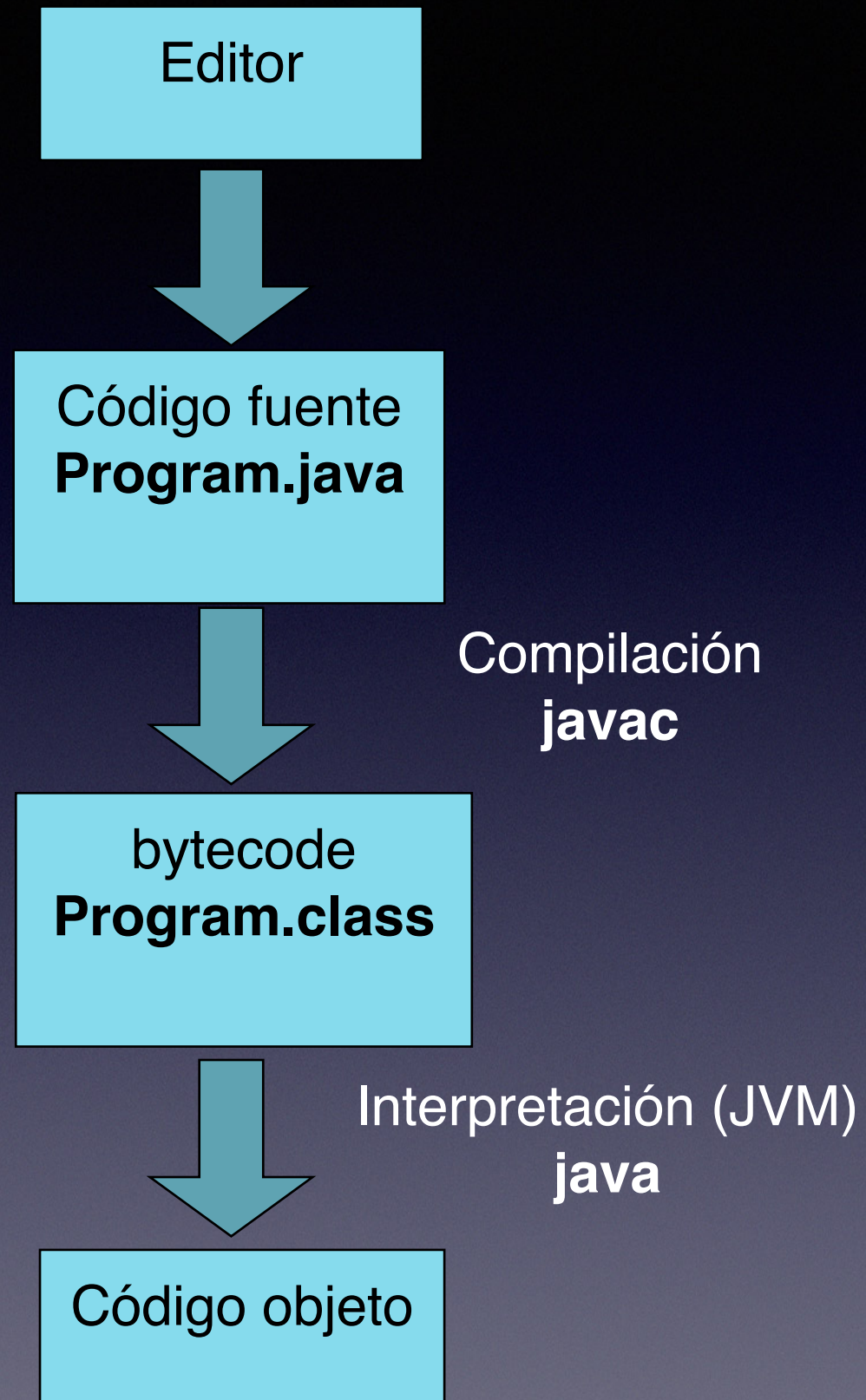
Sin punteros (manejados por el programador)

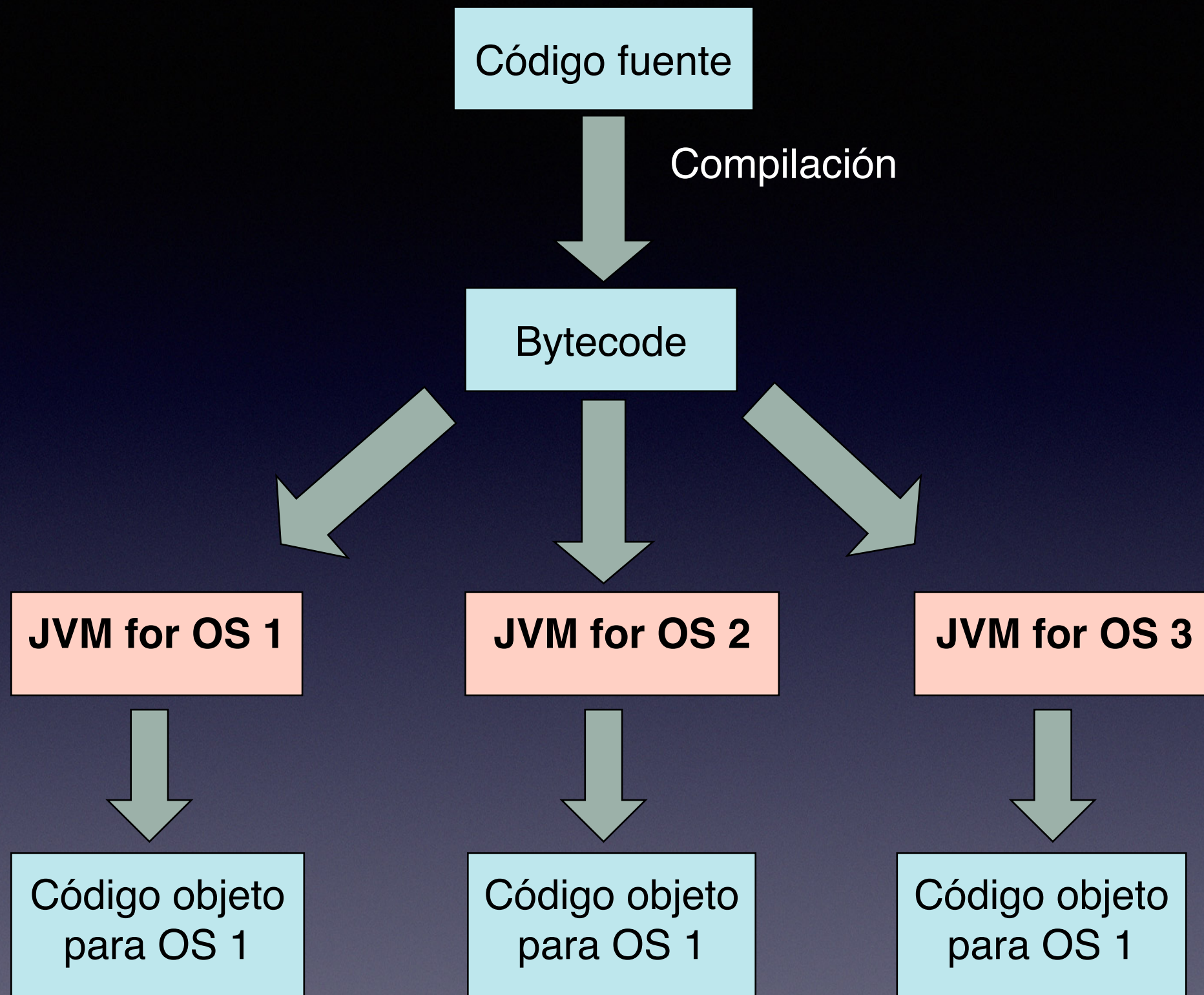
Manejo sencillo de memoria (recolector de basura)

Genera programas con pocos errores

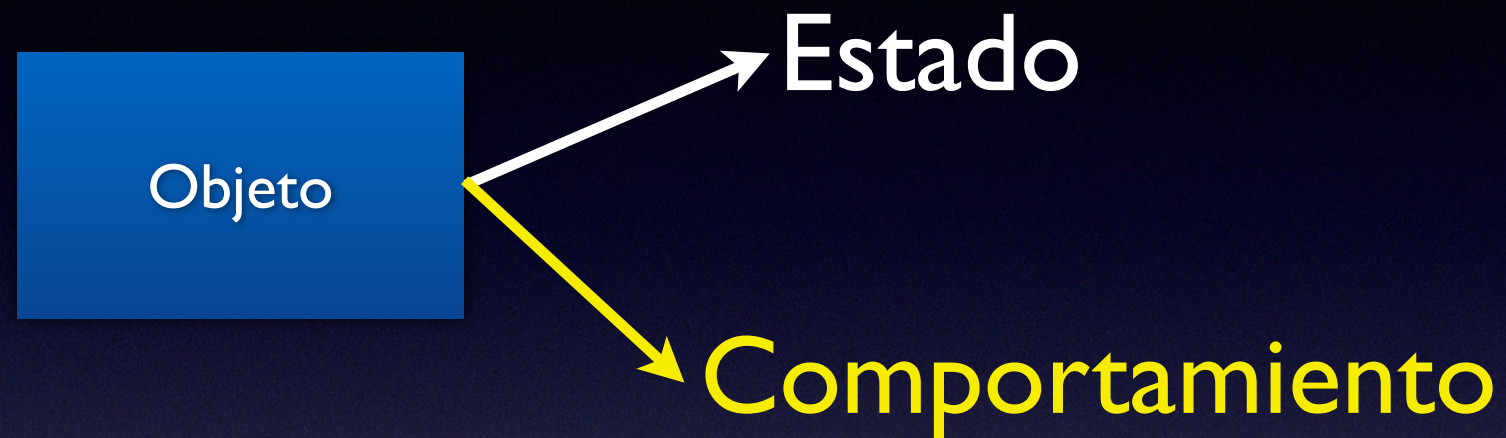
Multithilo



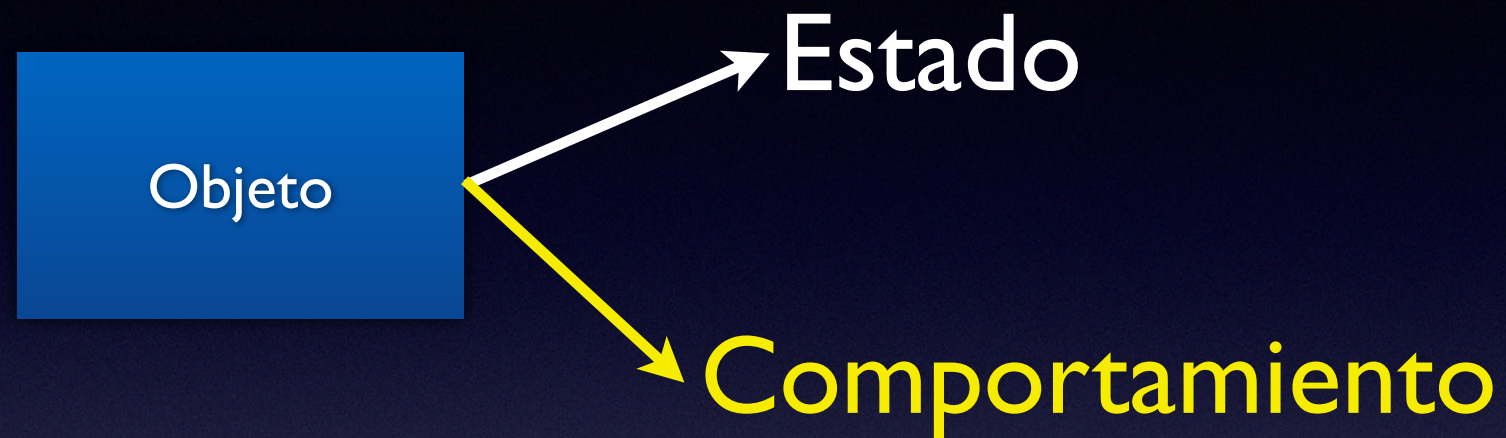




Conceptos POO



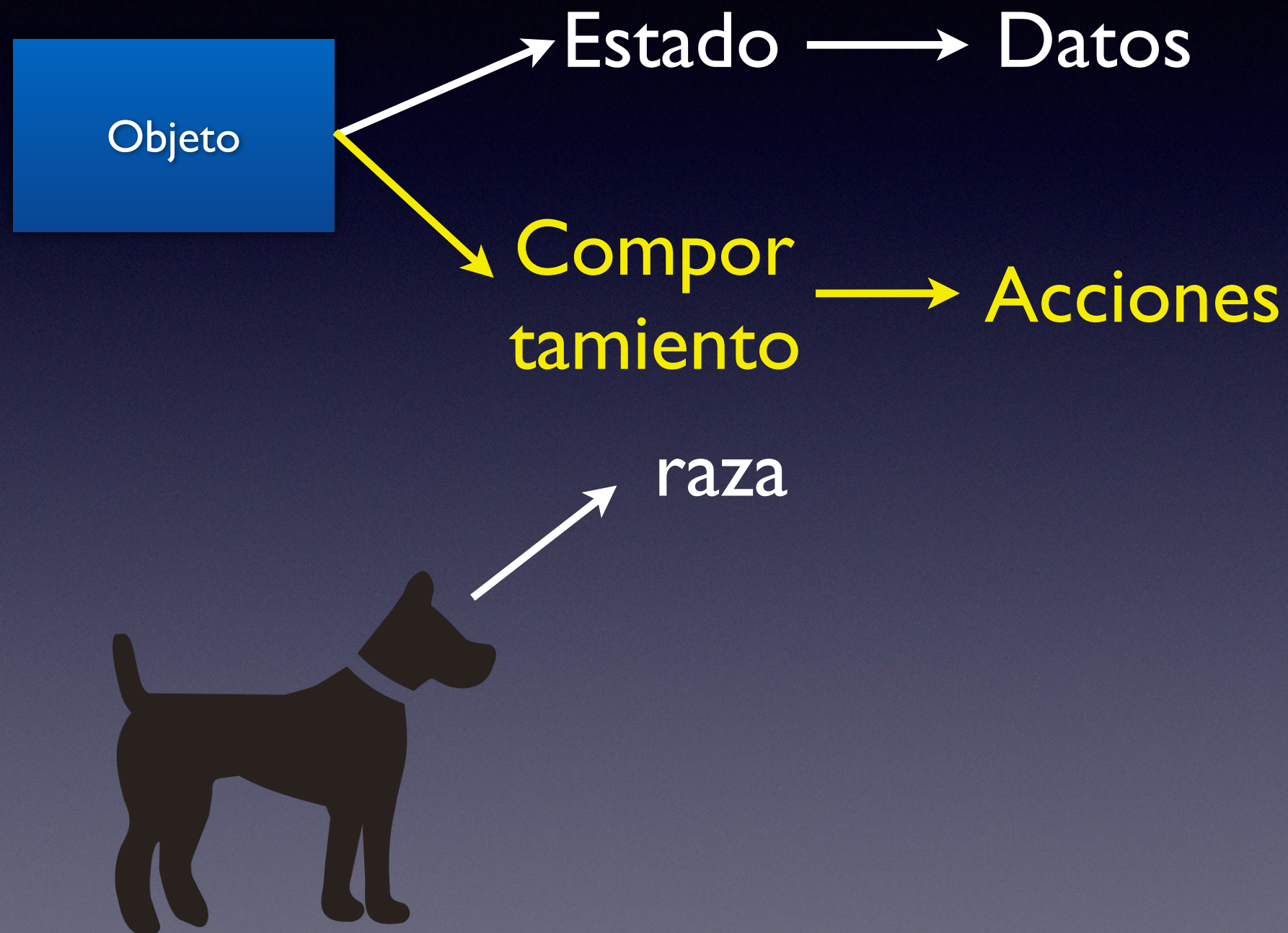
Conceptos POO



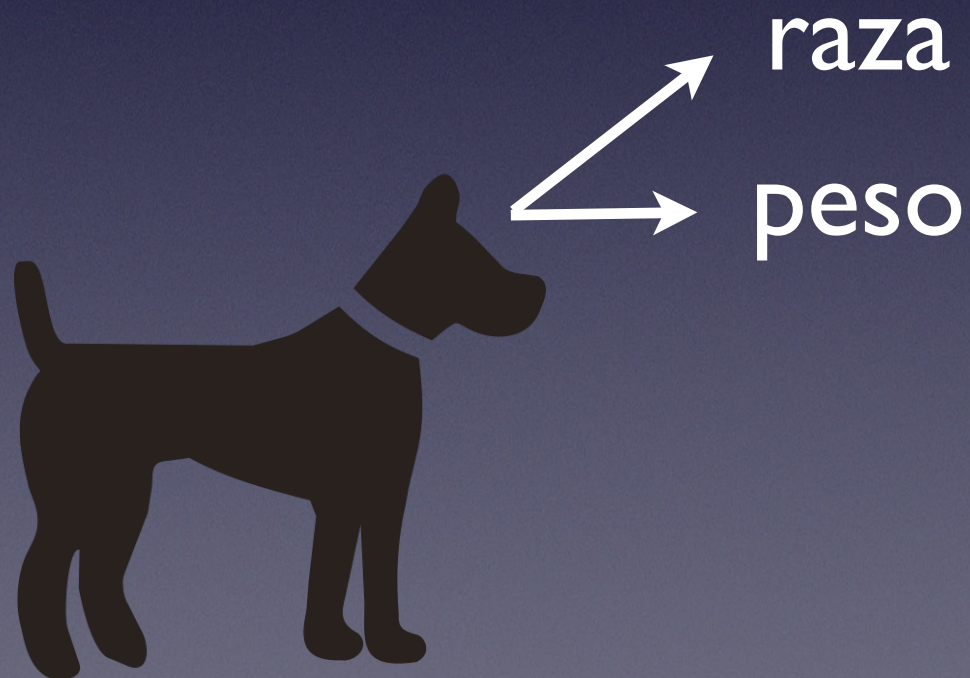
Conceptos POO



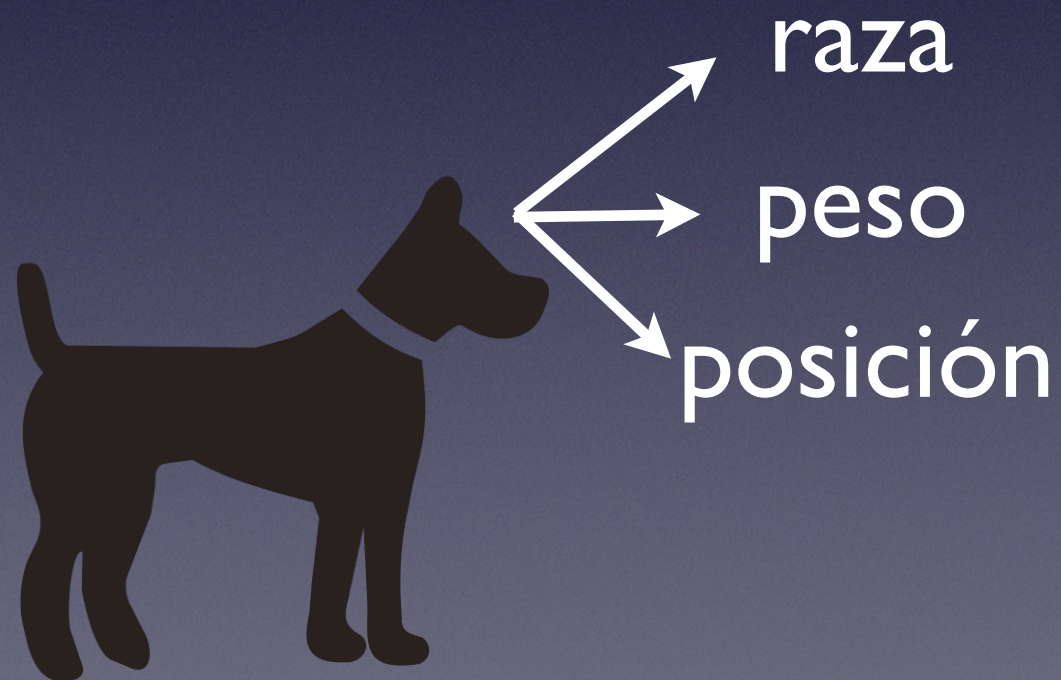
Conceptos POO



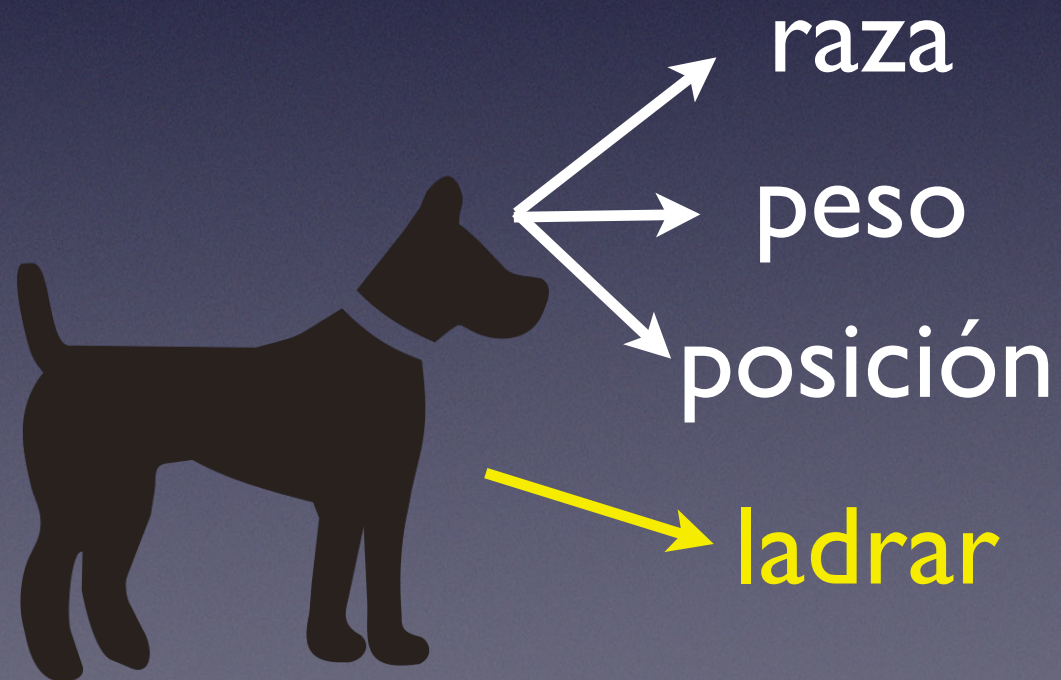
Conceptos POO



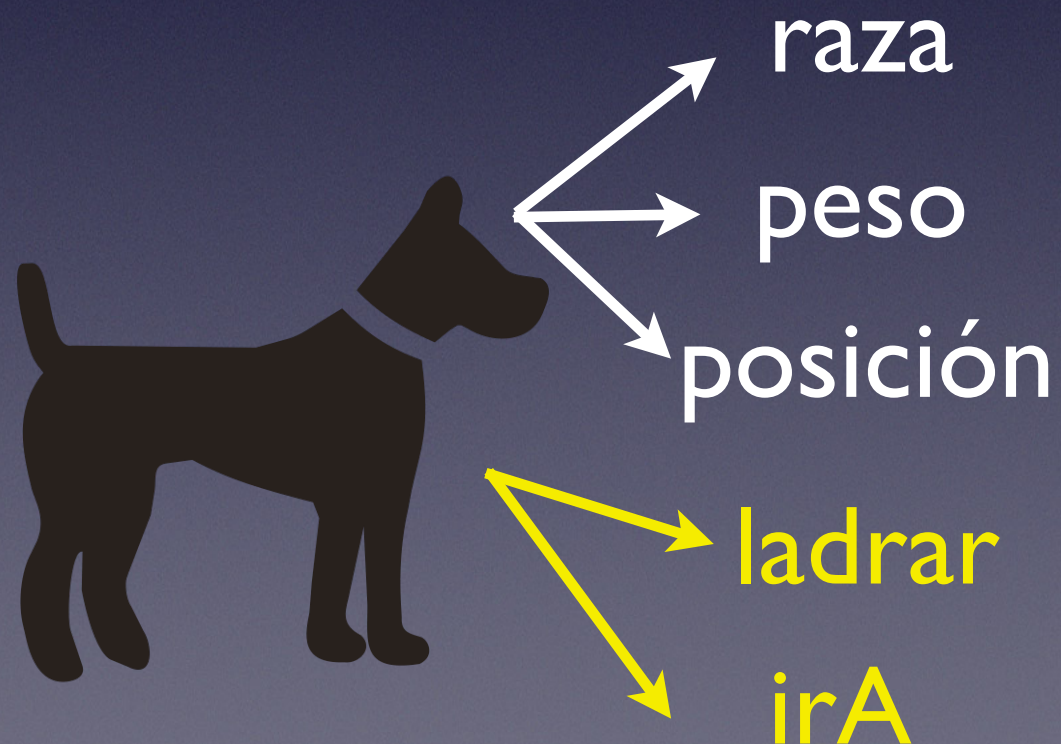
Conceptos POO



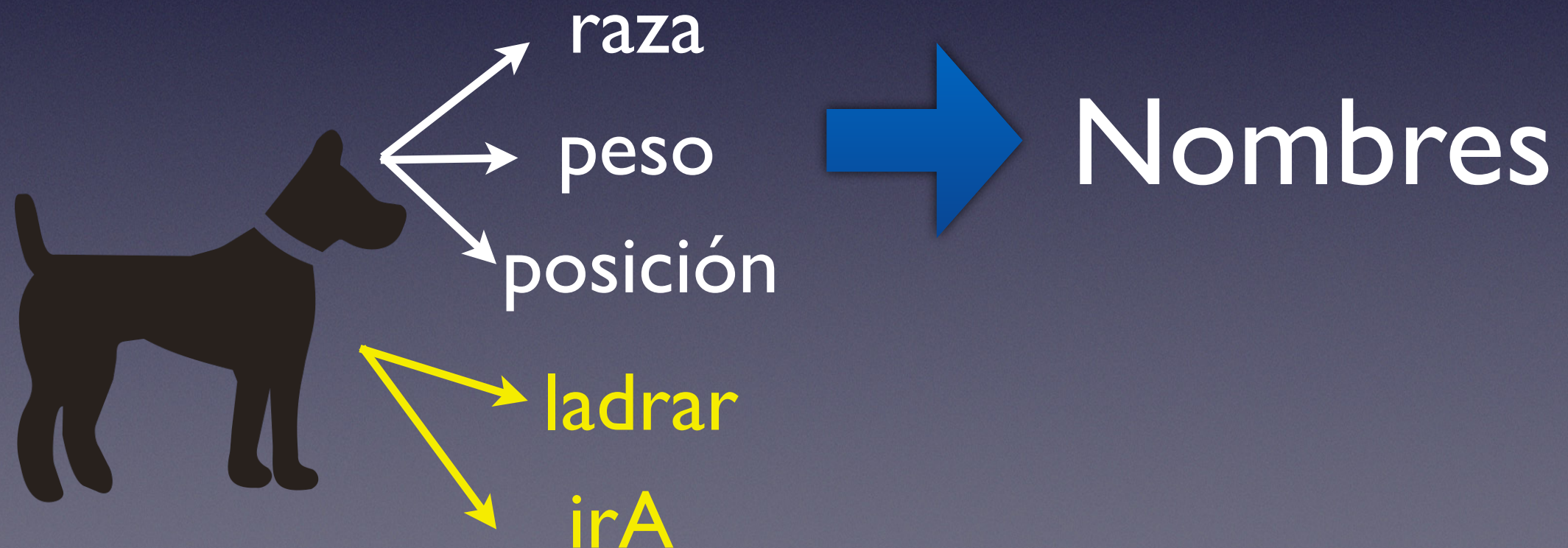
Conceptos POO



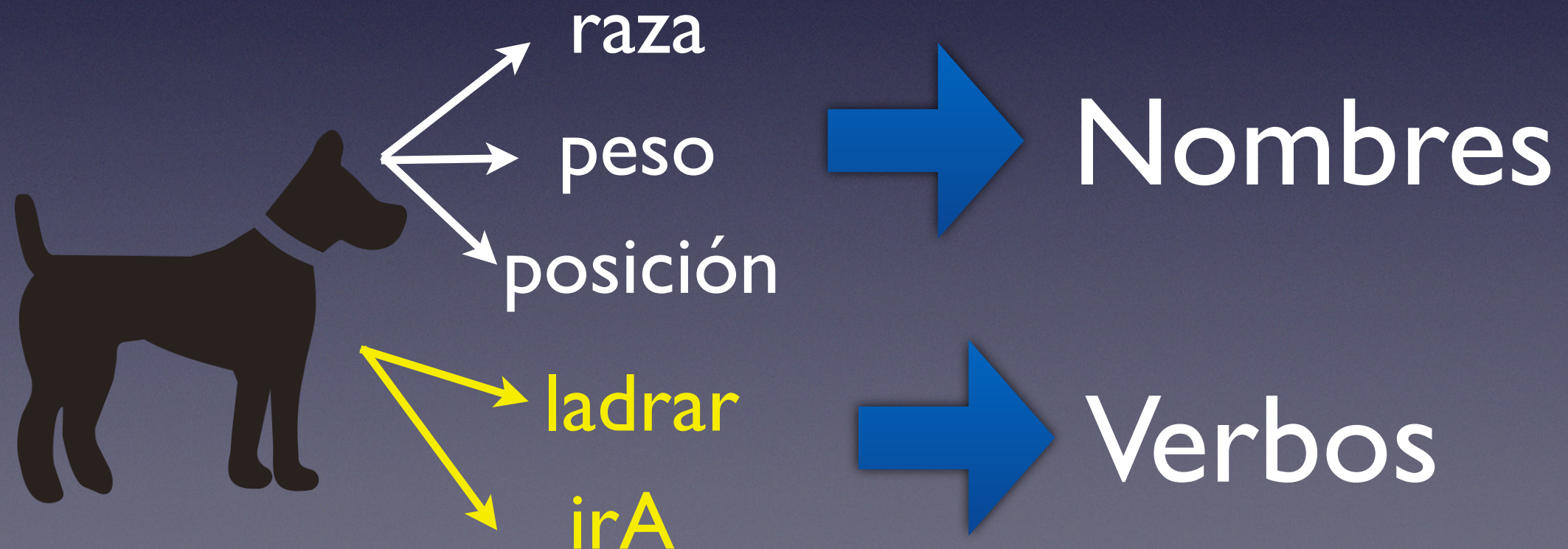
Conceptos POO



Conceptos POO



Conceptos POO



Conceptos POO



Conceptos POO

En Java



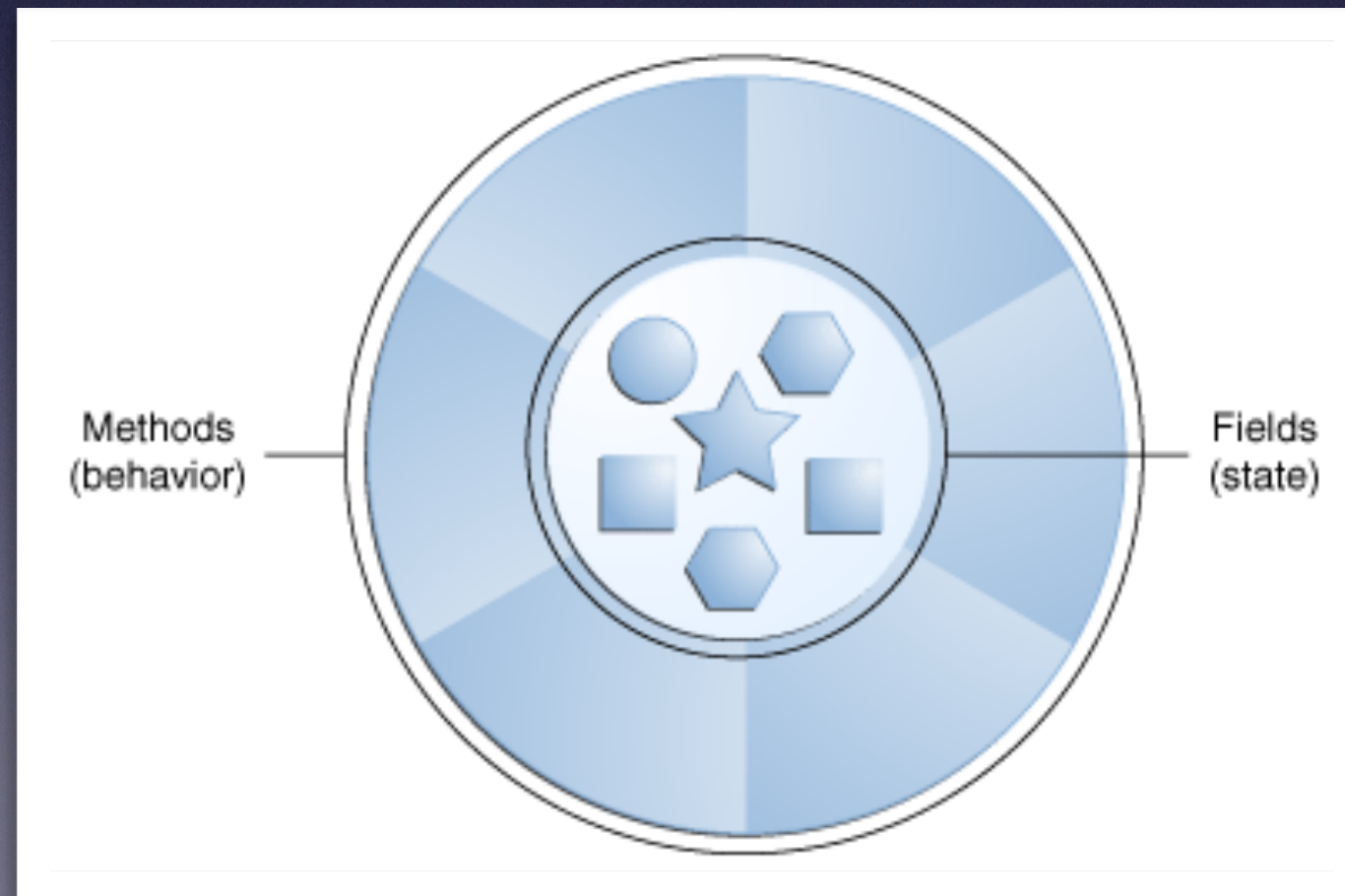
Conceptos POO

En Java



Conceptos POO

En Java

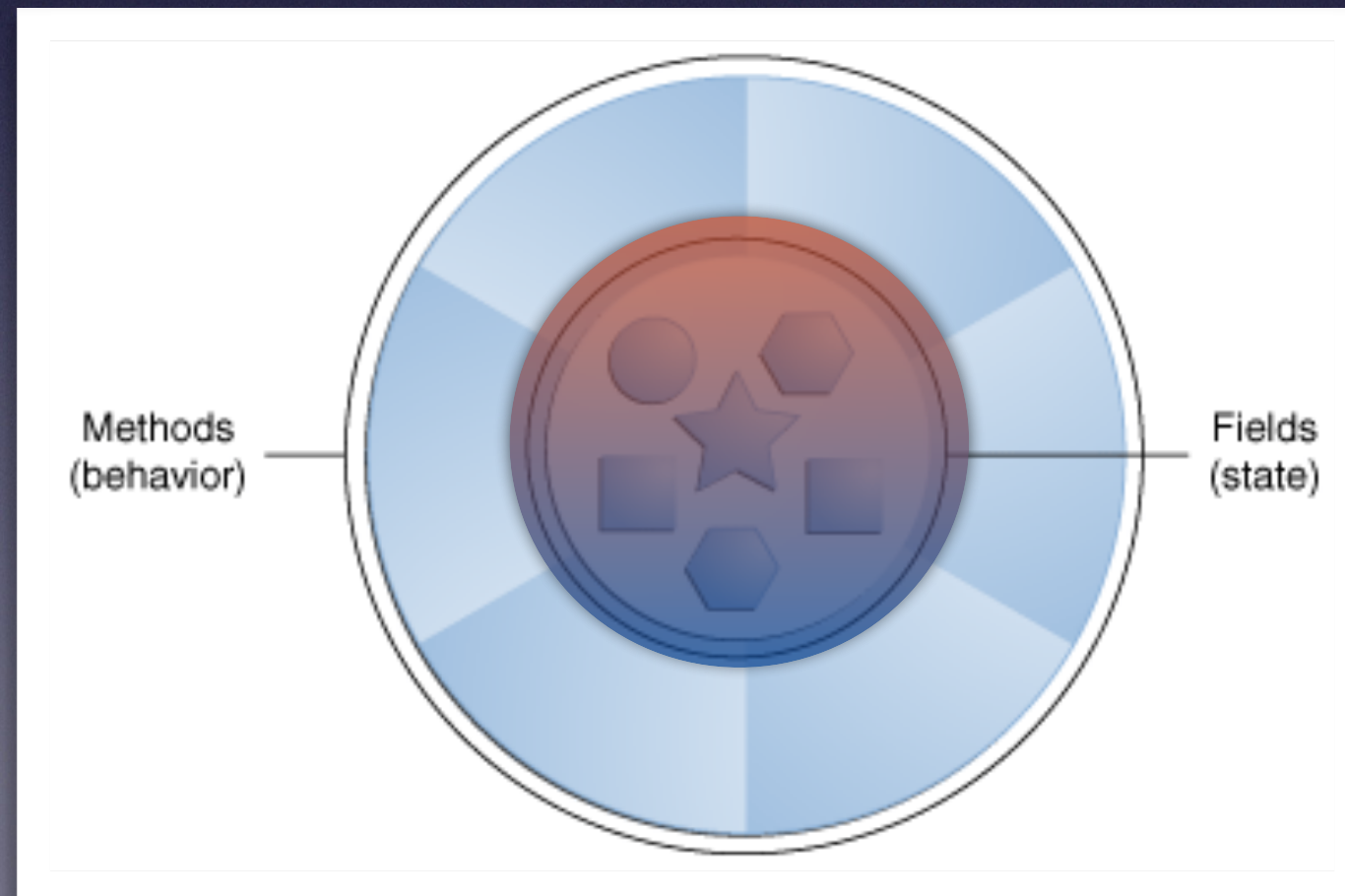


Conceptos POO

En Java

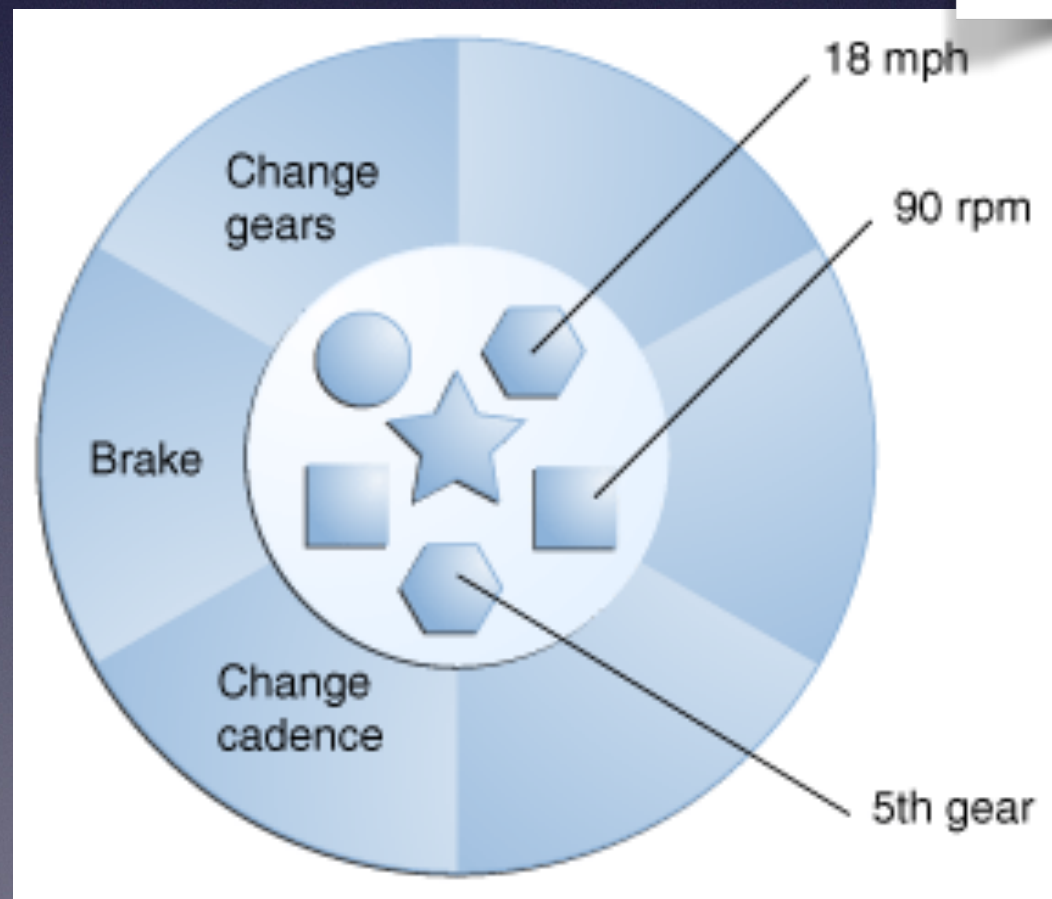


Encapsulación
de datos

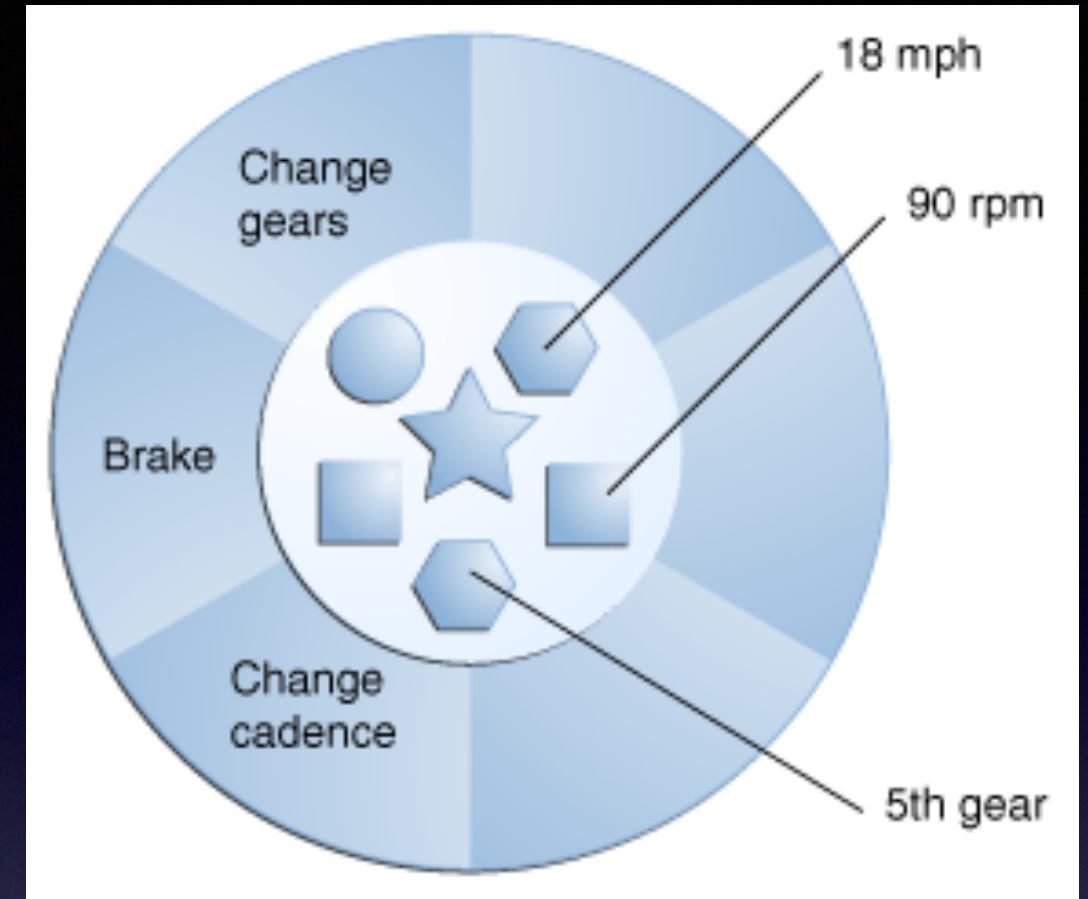


Conceptos POO

En Java

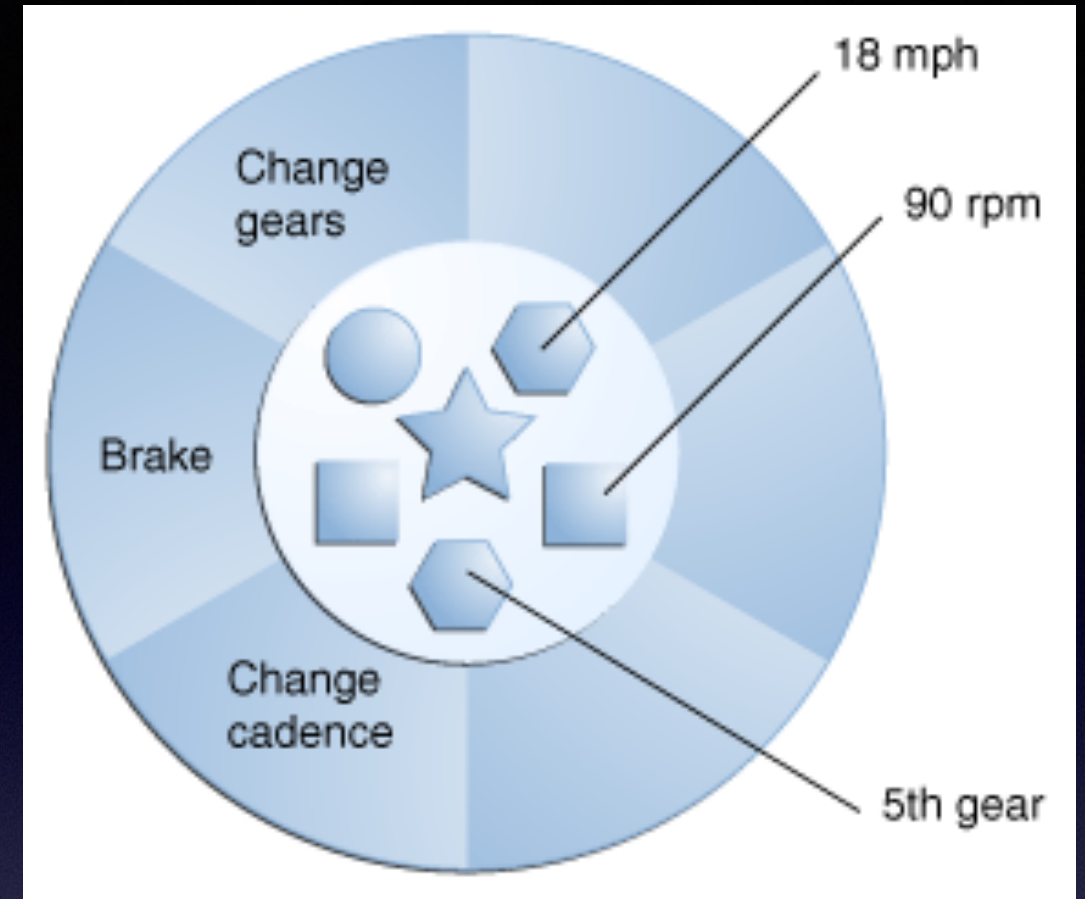


Ventajas POO



Ventajas POO

Modularidad

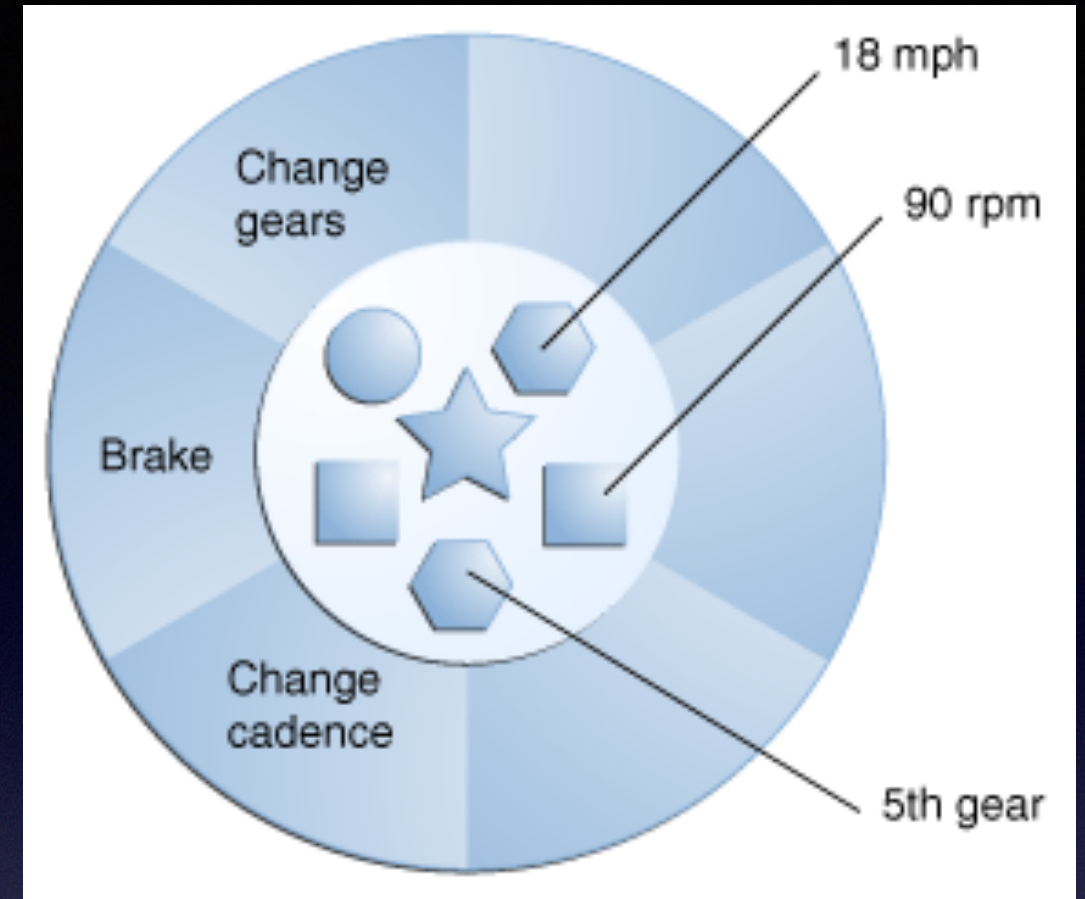
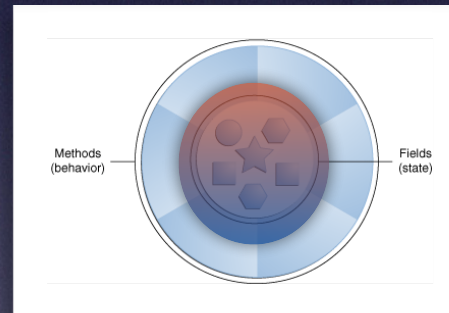


Ventajas POO

Modularidad

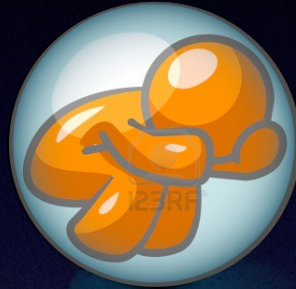


Ocultación de
información



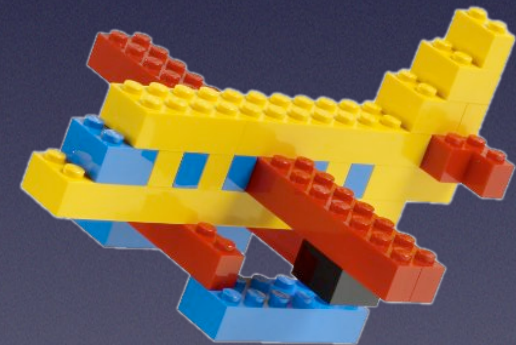
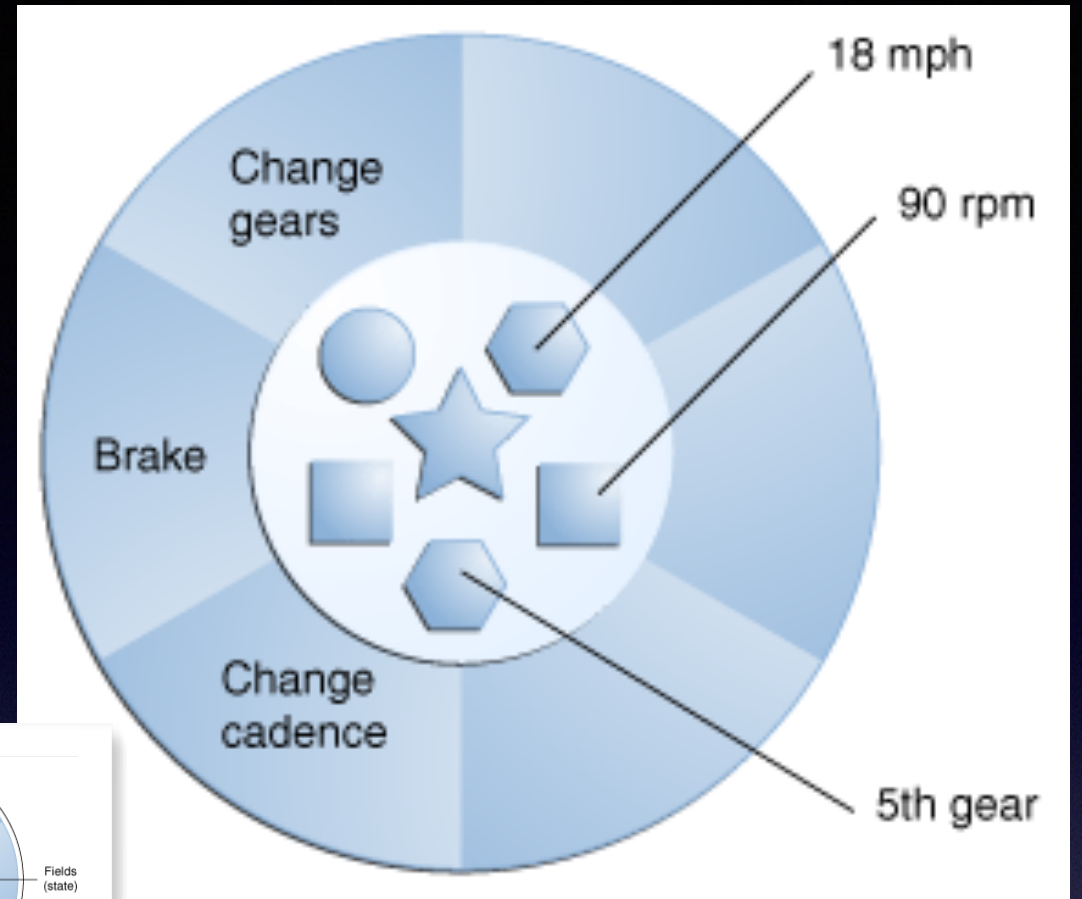
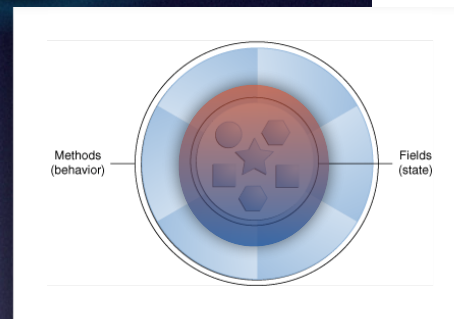
Ventajas POO

Modularidad



Ocultación de
información

Reutilización de código

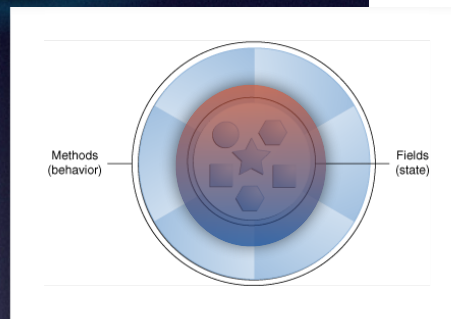


OOP advantages

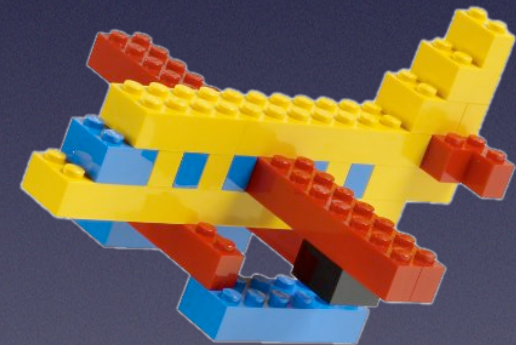
Modularity



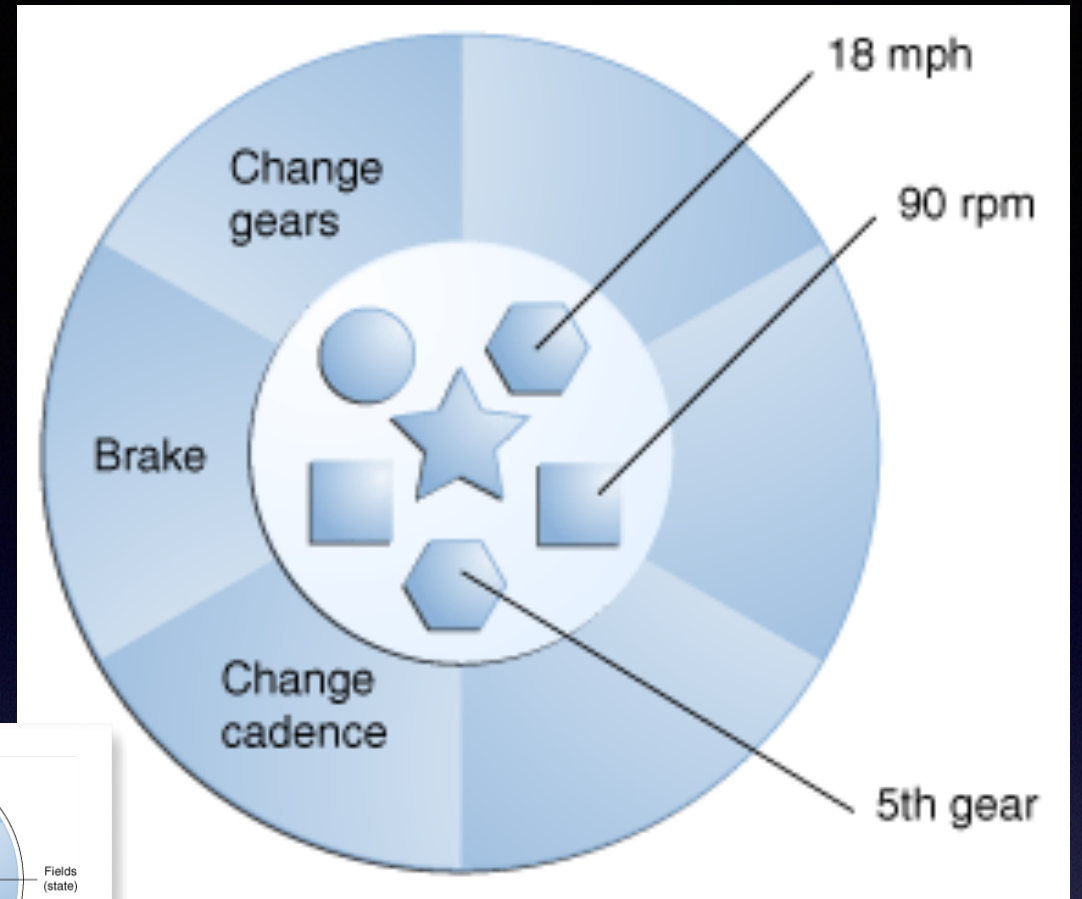
Information-hiding



Code re-use



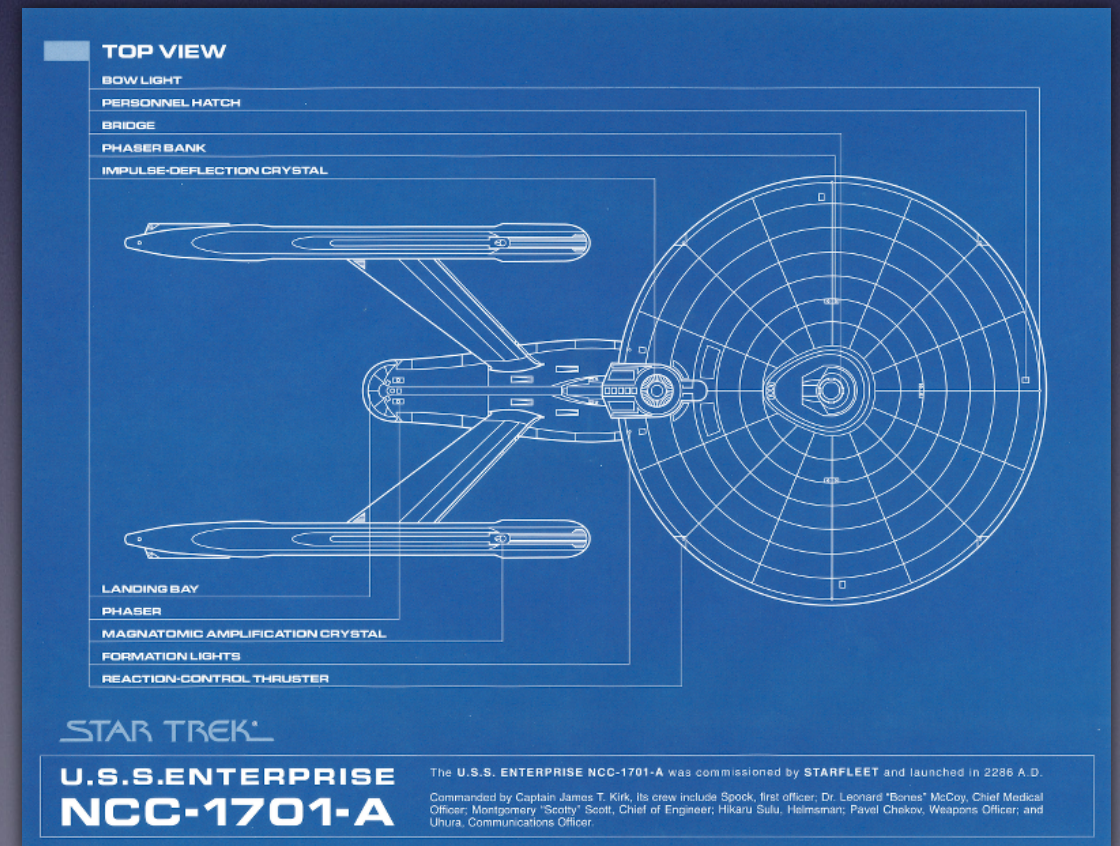
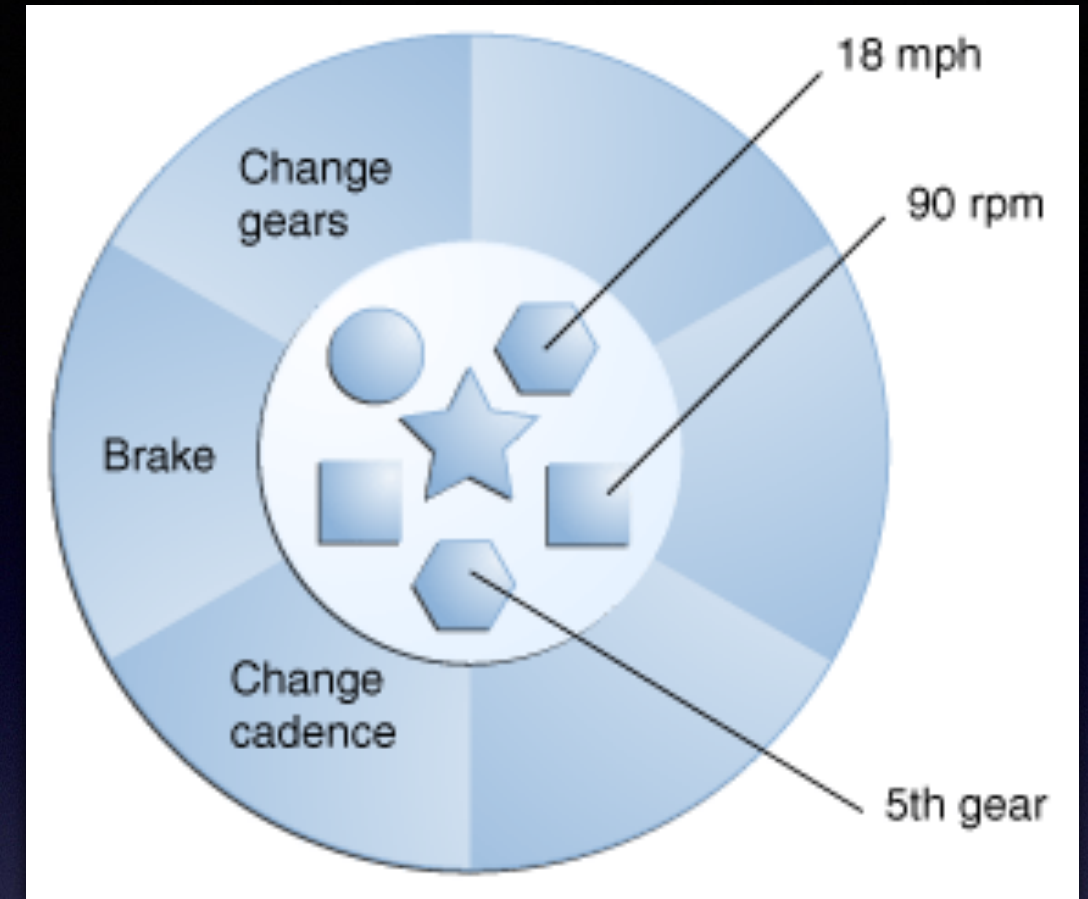
Pluggability and debugging ease



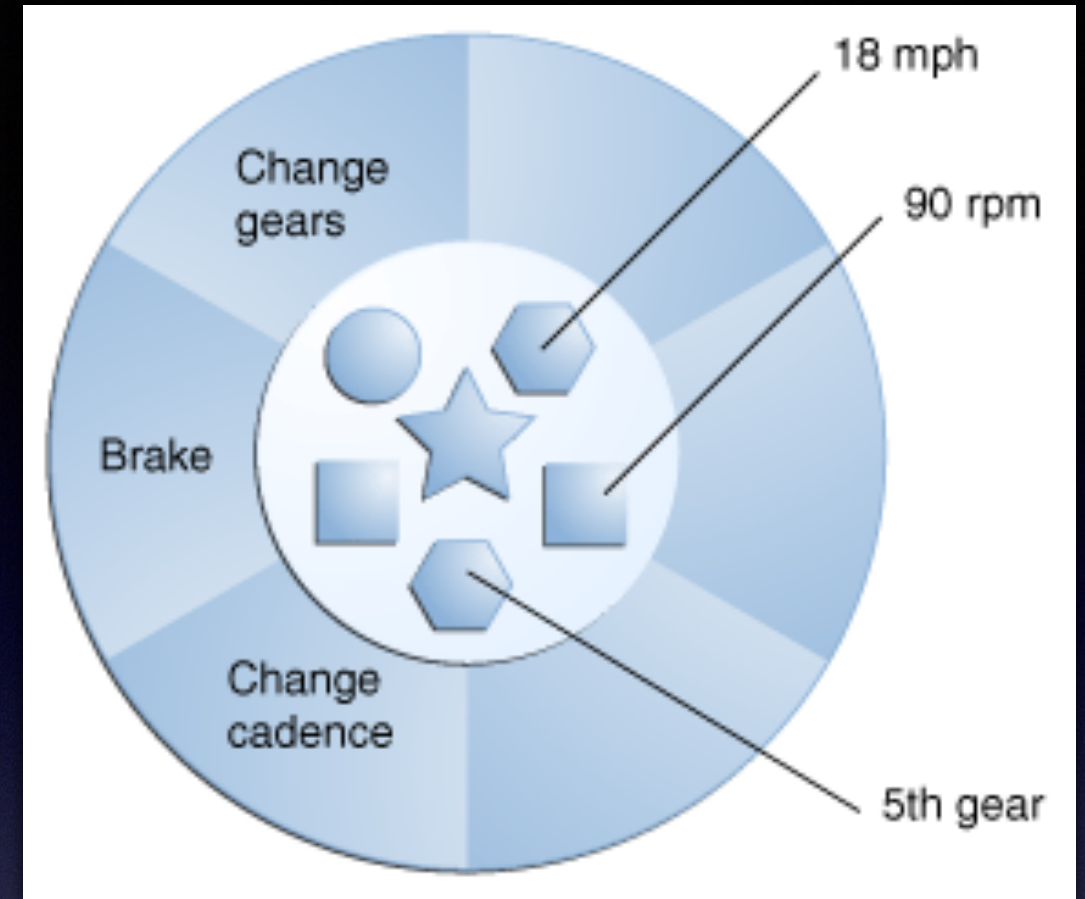
Una Clase



Como un molde o un
plano

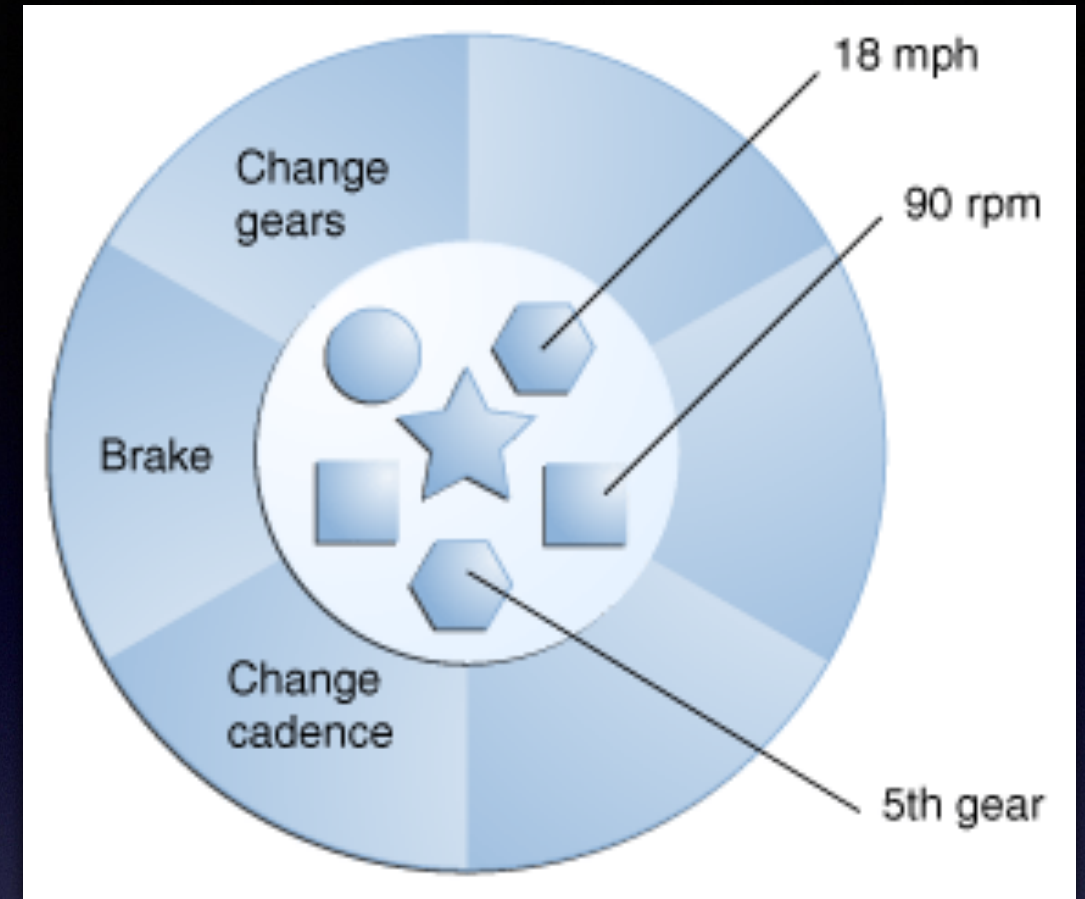


Una Clase



En términos orientado a objetos,
decimos que tu bicicleta es una
instancia de la *clase* Bicycle

Una Clase



En términos orientado a objetos,
decimos que tu bicicleta es una
instancia de la *clase* Bicycle

Instancia = Objeto

```
class Bicycle {
    int cadence = 0;
    int speed = 0;
    int gear = 1;

    void changeCadence(int newValue) {
        cadence = newValue;
    }

    void changeGear(int newValue) {
        gear = newValue;
    }

    void speedUp(int increment) {
        speed = speed + increment;
    }

    void applyBrakes(int decrement) {
        speed = speed - decrement;
    }

    void printStates() {
        System.out.println("cadence:" +
            cadence + " speed:" +
            speed + " gear:" + gear);
    }
}
```

Ejercicio I

Crea un Proyecto Bicycles Y añade esta clase Bicycle

```

class Bicycle {
    int cadence = 0;
    int speed = 0;
    int gear = 1;

    void changeCadence(int newValue) {
        cadence = newValue;
    }

    void changeGear(int newValue) {
        gear = newValue;
    }

    void speedUp(int increment) {
        speed = speed + increment;
    }

    void applyBrakes(int decrement) {
        speed = speed - decrement;
    }

    void printStates() {
        System.out.println("cadence:" +
            cadence + " speed:" +
            speed + " gear:" + gear);
    }
}

```

```

class BicycleDemo {
    public static void main(String[] args) {

        // Create two different
        // Bicycle objects
        Bicycle bike1 = new Bicycle();
        Bicycle bike2 = new Bicycle();

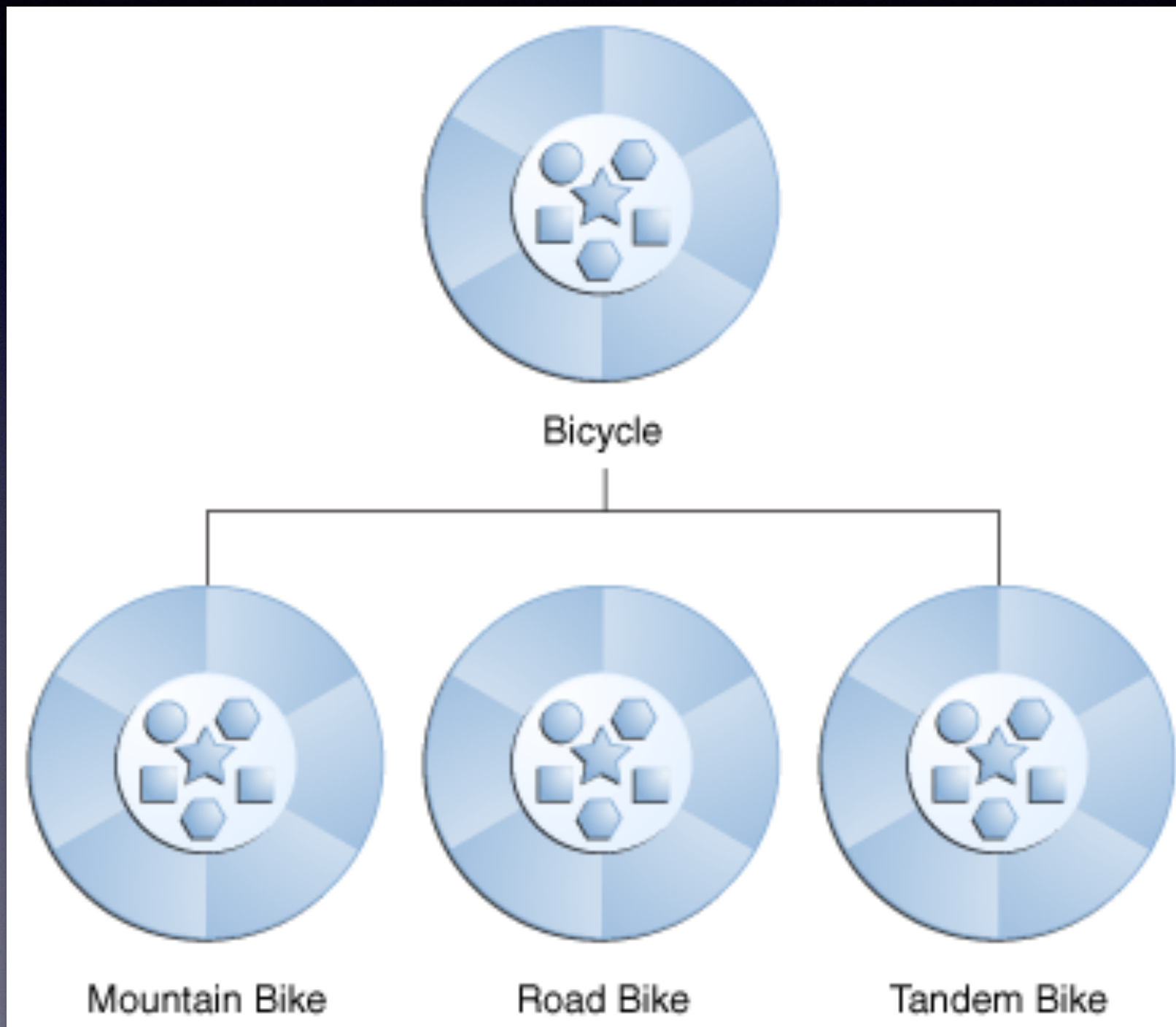
        // Invoke methods on
        // those objects
        bike1.changeCadence(50);
        bike1.speedUp(10);
        bike1.changeGear(2);
        bike1.printStates();

        bike2.changeCadence(50);
        bike2.speedUp(10);
        bike2.changeGear(2);
        bike2.changeCadence(40);
        bike2.speedUp(10);
        bike2.changeGear(3);
        bike2.printStates();
    }
}

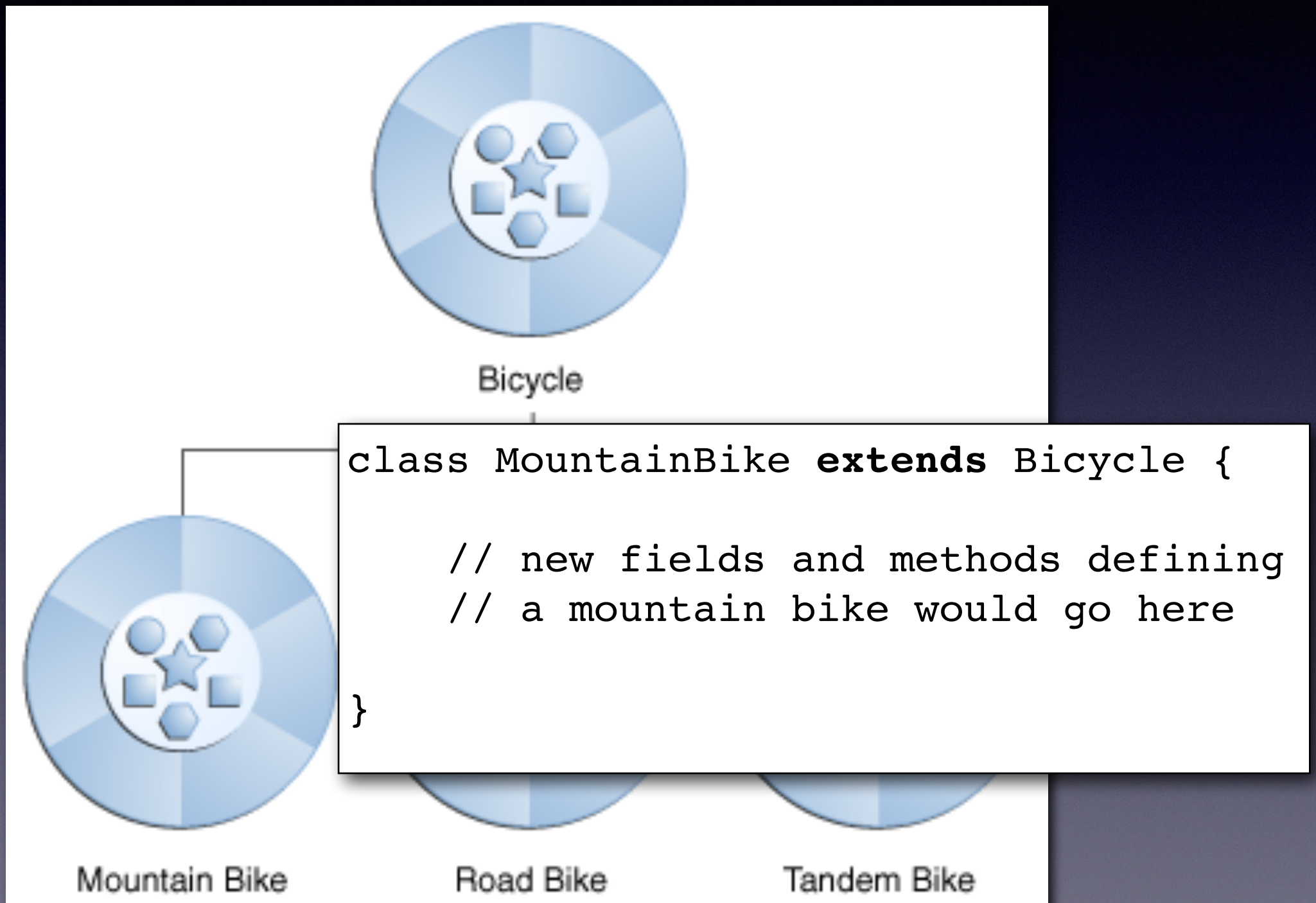
```

Luego añade esta clase BicycleDemo en un nuevo
 fichero .java y comparte el proyecto en Github

Herencia



Herencia



Herencia



```
class TandemBike extends Bicycle {  
  
    int numSeats = 2;  
  
    void applyBrakes(int decrement1, int decrement2) {  
        speed = speed - decrement1 - decrement2;  
        if (speed < 0) {  
            speed=0;  
        }  
    }  
}
```



Mountain Bike



Road Bike



Tandem Bike

Herencia

```
class TandemBike extends Bicycle {  
  
    int numSeats = 2;  
  
    void applyBrakes(int decrement1, int decrement2) {  
        speed = speed - decrement1 - decrement2;  
        if (speed < 0) {  
            speed = 0;  
        }  
    }  
}
```

Ejercicio 2

Añade la clase TandemBike al proyecto del Ejercicio 1. Haz un commit (menu:VCS / Commit ...) y sube el cmmid al repositorio remote Github (menu:VCS / Git / Push)

Her

```
class TandemBike extends Bicycle {  
  
    int numSeats = 2;  
  
    void applyBrakes(int deceleration)  
    {  
        speed = speed - deceleration;  
        if (speed < 0) {  
            speed = 0;  
        }  
    }  
}
```

```
class BicycleDemo2 {  
    public static void main(String[] args) {  
  
        // Create two different  
        // Bicycle objects  
        Bicycle bike1 = new Bicycle();  
        TandemBike bike2 = new TandemBike();  
  
        // Invoke methods on  
        // those objects  
        bike1.changeCadence(50);  
        bike1.speedUp(10);  
        bike1.changeGear(2);  
        bike1.printStates();  
  
        bike2.changeCadence(50);  
        bike2.speedUp(10);  
        bike2.changeGear(2);  
        bike2.changeCadence(40);  
        bike2.speedUp(10);  
        bike2.changeGear(3);  
        bike2.applyBrakes(5,3);  
        bike2.printStates();  
    }  
}
```

Mountain Bike

Road Bike

Tandem Bike

Her

```
class TandemBike extends Bi

    int numSeats = 2;

    void applyBrakes(int de
        speed = speed - de
        if (speed < 0) {
            speed=0;
        }
    }
}
```

```
class BicycleDemo2 {
    public static void main(String[] args) {

        // Create two different
        // Bicycle objects
        Bicycle bike1 = new Bicycle();
        TandemBike bike2 = new TandemBike();

        // Invoke methods on
        // those objects
        bike1.changeCadence(50);
        bike1.speedUp(10);
        bike1.changeGear(2);
        bike1.printStates();

        bike2.changeCadence(50);
        bike2.speedUp(10);
        bike2.changeGear(2);
        bike2.changeCadence(40);
        bike2.speedUp(10);
        bike2.changeGear(3);
        bike2.applyBrakes(5,3);
        bike2.applyBrakes(2);
        bike2.printStates();
    }
}
```

Mountain Bike

Road Bike

Tandem Bike

Her

Ejercicio 2

Añade la clase `BicycleDemo2` a tu proyecto. Haz un commit (menu:VCS / Commit ...) luego sobre el commit al repositorio remoto en Github (menu:VCS / Git / Push)

```
class BicycleDemo2 {
    public static void main(String[] args) {

        // Create two different
        // Bicycle objects
        Bicycle bike1 = new Bicycle();
        TandemBike bike2 = new TandemBike();

        // Invoke methods on
        // those objects
        bike1.changeCadence(50);
        bike1.speedUp(10);
        bike1.changeGear(2);
        bike1.printStates();

        bike2.changeCadence(50);
        bike2.speedUp(10);
        bike2.changeGear(2);
        bike2.changeCadence(40);
        bike2.speedUp(10);
        bike2.changeGear(3);
        bike2.applyBrakes(5,3);
        bike2.applyBrakes(2);
        bike2.printStates();

    }
}
```

Her

```
class BicycleDemo2 {  
    public static void main(String[] args) {  
  
        // Create two different  
        // Bicycle objects  
        Bicycle bike1 = new Bicycle();  
        TandemBike bike2 = new TandemBike();  
    }  
}
```

```
class TandemBike extends Bicycle {  
  
    int numSeats=2;  
  
    void applyBrakes(int decrement1, int decrement2) {  
        speed = speed - decrement1 - decrement2;  
        if (speed < 0) {  
            speed = 0;  
        }  
    }  
  
    void printStates() {  
        super.printStates();  
        System.out.println("num Seats:" + numSeats);  
    }  
}
```

Herencia

```
class TandemBike extends Bicycle {  
  
    int numSeats=2;  
  
    void applyBrakes(int decrement1, int decrement2) {  
        speed = speed - decrement1 - decrement2;  
        if (speed < 0) {  
            speed = 0;  
        }  
    }  
    void printStates() {  
        super.printStates();  
        System.out.println("num Seats:" + numSeats);  
    }  
}
```

Exercise 2

Modificar la clase TandemBike. Haz commit (menu:VCS / Commit ...) y sube el commit al repositorio remoto en Github (menu:VCS / Git / Push)

Her

```
class TandemBike extends Bicycle {  
  
    int numSeats = 2;  
  
    void applyBrakes(int decrease) {  
        speed = speed - decrease;  
        if (speed < 0) {  
            speed = 0;  
        }  
    }  
  
    void printStates() {  
        super.printStates();  
        System.out.println("Tandem Bike");  
    }  
}
```

```
class BicycleDemo2 {  
    public static void main(String[] args) {  
  
        // Create two different  
        // Bicycle objects  
        Bicycle bike1 = new Bicycle();  
        TandemBike bike2 = new TandemBike();  
  
        // Invoke methods on  
        // those objects  
        bike1.changeCadence(50);  
        bike1.speedUp(10);  
        bike1.changeGear(2);  
        bike1.printStates();  
  
        bike2.changeCadence(50);  
        bike2.speedUp(10);  
        bike2.changeGear(2);  
        bike2.changeCadence(40);  
        bike2.speedUp(10);  
        bike2.changeGear(3);  
        bike2.applyBrakes(5,3);  
        bike2.applyBrakes(2);  
        bike2.printStates();  
    }  
}
```

Mountain Bike

Road Bike

Tandem Bike

Clases de variables

- Atributos. (Variables de instancia)
- Atributos estáticos. (Variables de clase)
- Variables locales
- Parámetros

Clases de

- Atributos. (Variables)
- Atributos estáticos.
- Variables locales
- Parámetros

```
class Bicycle {
    int cadence = 0;
    int speed = 0;
    int gear = 1;
    static int numBikes = 0;
    static final MAX_GEAR = 8;

    void changeCadence(int newValue) {
        cadence = newValue;
    }

    void changeGear(int newValue) {
        int i = 0;
        ...
        gear = newValue;
    }

    void speedUp(int increment) {
        speed = speed + increment;
    }

    void applyBrakes(int decrement) {
        speed = speed - decrement;
    }

    void printStates() {
        System.out.println("cadence:" +
```

Clases de

- **Atributos.** (Variables
- Atributos estáticos.
- Variables locales
- Parámetros

```
class Bicycle {  
    int cadence = 0;  
    int speed = 0;  
    int gear = 1;  
    static int numBikes = 0;  
    static final MAX_GEAR = 8;  
  
    void changeCadence(int newValue) {  
        cadence = newValue;  
    }  
  
    void changeGear(int newValue) {  
        int i = 0;  
        ...  
        gear = newValue;  
    }  
  
    void speedUp(int increment) {  
        speed = speed + increment;  
    }  
  
    void applyBrakes(int decrement) {  
        speed = speed - decrement;  
    }  
  
    void printStates() {  
        System.out.println("cadence:" +
```

Clases de

- Atributos. (Variables)
- Atributos estáticos.
- Variables locales
- Parámetros

```
class Bicycle {
    int cadence = 0;
    int speed = 0;
    int gear = 1;
    static int numBikes = 0;
    static final MAX_GEAR = 8;

    void changeCadence(int newValue) {
        cadence = newValue;
    }

    void changeGear(int newValue) {
        int i = 0;
        ...
        gear = newValue;
    }

    void speedUp(int increment) {
        speed = speed + increment;
    }

    void applyBrakes(int decrement) {
        speed = speed - decrement;
    }

    void printStates() {
        System.out.println("cadence:" +
```

Clases de

- Atributos. (Variables)
- Atributos estáticos.
- Variables locales
- Parámetros

```
class Bicycle {
    int cadence = 0;
    int speed = 0;
    int gear = 1;
    static int numBikes = 0;
    static final MAX_GEAR = 8;

    void changeCadence(int newValue) {
        cadence = newValue;
    }

    void changeGear(int newValue) {
        int i = 0;
        ...
        gear = newValue;
    }

    void speedUp(int increment) {
        speed = speed + increment;
    }

    void applyBrakes(int decrement) {
        speed = speed - decrement;
    }

    void printStates() {
        System.out.println("cadence: " +
```

Clases de

- Atributos. (Variables)
- Atributos estáticos.
- Variables locales
- Parámetros

```
class Bicycle {  
    int cadence = 0;  
    int speed = 0;  
    int gear = 1;  
    static int numBikes = 0;  
    static final MAX_GEAR = 8;  
  
    void changeCadence(int newValue) {  
        cadence = newValue;  
    }  
  
    void changeGear(int newValue) {  
        int i = 0;  
        ...  
        gear = newValue;  
    }  
  
    void speedUp(int increment) {  
        speed = speed + increment;  
    }  
  
    void applyBrakes(int decrement) {  
        speed = speed - decrement;  
    }  
  
    void printStates() {  
        System.out.println("cadence: " +
```

Tipos en Java

Tiempo simple	Qué almacena	Rango
byte	enteros de 8 bits	-128 to 127
short	enteros de 16 bits	-32,768 to 32,767
int	enteros de 32 bits	-2,147,483,648 to 2,147,483,647
long	enteros de 64 bits	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	Reales en coma flotante de 32 bits	6 significative digits (10^{-46} to 10^{38})
double	Reales en coma flotante de 64 bits	15 significative digits (10^{-324} to 10^{308})
char	Carácter unicode de 16 bits	
boolean	Variable booleana	true / false

Tipos en Java

Tipo de datos	Valor por defecto (atributos)
byte	0
short	0
int	0
long	0L
float	0.0f
double	0.0d
char	'\u0000'
String (or any object)	null
boolean	false

Literales

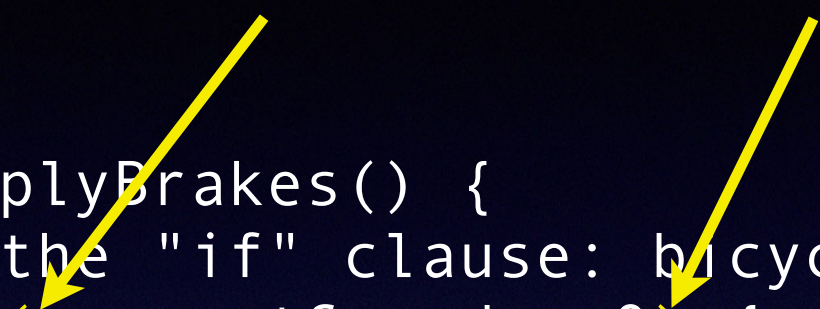
```
boolean result = true;  
char capitalC = 'C';  
byte b = 100;  
short s = 10000;  
int i = 100000;  
// The number 26, in decimal  
int decVal = 26;  
// The number 26, in hexadecimal  
int hexVal = 0x1a;  
double d1 = 123.4;  
// same value as d1, but in  
scientific notation  
double d2 = 1.234e2;  
float f1 = 123.4f;  
String name = "V́ctor";
```

if

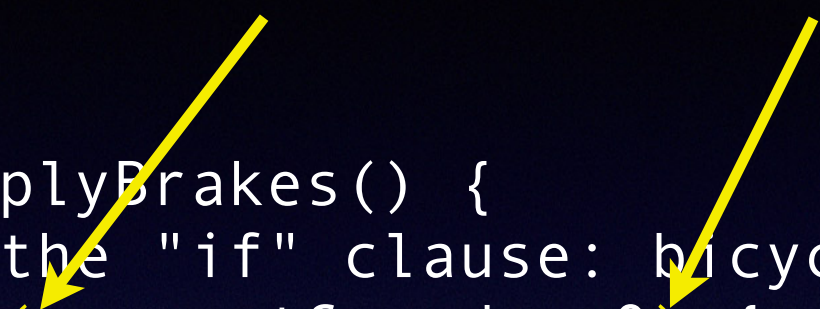
```
void applyBrakes() {  
    // the "if" clause: bicycle must be moving  
    if (currentSpeed > 0) {  
        // the "then" clause: decrease current speed  
        currentSpeed--;  
    }  
}
```

if

```
void applyBrakes() {  
    // the "if" clause: bicycle must be moving  
    if (currentSpeed > 0) {  
        // the "then" clause: decrease current speed  
        currentSpeed--;  
    }  
}
```



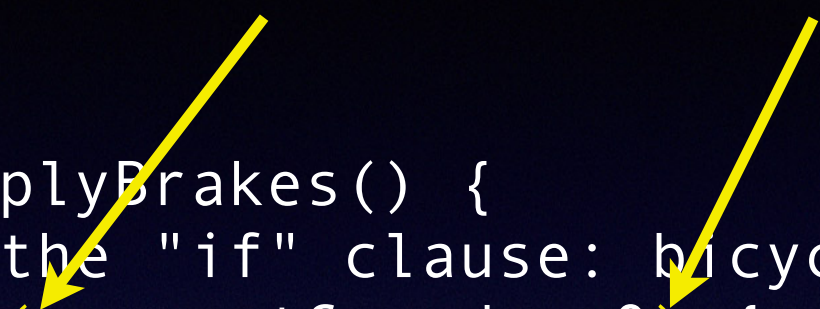
if



```
void applyBrakes() {  
    // the "if" clause: bicycle must be moving  
    if (currentSpeed > 0) {  
        // the "then" clause: decrease current speed  
        currentSpeed--;  
    }  
}
```

```
void applyBrakes() {  
    // same as above, but without braces  
    if (currentSpeed > 0)  
        currentSpeed--;  
}
```

if



```
void applyBrakes() {  
    // the "if" clause: bicycle must be moving  
    if (currentSpeed > 0) {  
        // the "then" clause: decrease current speed  
        currentSpeed--;  
    }  
}
```

```
void applyBrakes() {  
    // same as above, but without braces  
    if (currentSpeed > 0)  
        currentSpeed--;  
}
```

if-else

```
void applyBrakes() {  
    // the "if" clause: bicycle must be moving  
    if (currentSpeed > 0) {  
        // the "then" clause: decrease current speed  
        currentSpeed--;  
    } else {  
        System.err.println("The bicycle has already stopped!");  
    }  
}
```

if-else

```
void applyBrakes() {  
    // the "if" clause: bicycle must be moving  
    if (currentSpeed > 0) {  
        // the "then" clause: decrease current speed  
        currentSpeed--;  
    } else {  
        System.err.println("The bicycle has already stopped!");  
    }  
}
```

```
void applyBrakes() {  
    // same as above, but without braces  
    if (currentSpeed > 0)  
        currentSpeed--;  
    else  
        System.err.println("The bicycle has already stopped!");  
}
```

if-else

```
import java.util.Scanner;
class IfElseDemo {
    public static void main(String[] args) {

        int testScore;
        char grade;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        testScore = inputValue.nextInt();

        if (testScore >= 90) {
            grade = 'A';
        } else {
            if (testScore >= 80) {
                grade = 'B';
            } else {
                if (testScore >= 70) {
                    grade = 'C';
                } else {
                    if (testScore >= 60) {
                        grade = 'D';
                    } else {
                        grade = 'F';
                    }
                }
            }
        }

        System.out.println("Grade = " + grade);
    }
}
```

if-else

```
import java.util.Scanner;
class IfElseDemo {
    public static void main(String[] args) {

        int testScore;
        char grade;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        testScore = inputValue.nextInt();

        if (testScore >= 90) {
            grade = 'A';
        } else {
            if (testScore >= 80) {
                grade = 'B';
            } else {
                if (testScore >= 70) {
                    grade = 'C';
                } else {
                    if (testScore >= 60) {
                        grade = 'D';
                    } else {
                        grade = 'F';
                    }
                }
            }
        }

        System.out.println("Grade = " + grade);
    }
}
```

if-else

```
import java.util.Scanner;
class IfElseDemo {
    public static void main(String[] args) {

        int testScore;
        char grade;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        testScore = inputValue.nextInt();

        if (testScore >= 90) {
            grade = 'A';
        } else {
            if (testScore >= 80) {
                grade = 'B';
            } else {
                if (testScore >= 70) {
                    grade = 'C';
                } else {
                    if (testScore >= 60) {
                        grade = 'D';
                    } else {
                        grade = 'F';
                    }
                }
            }
        }

        System.out.println("Grade = " + grade);
    }
}
```

if-else

```
import java.util.Scanner;
class IfElseDemo {
    public static void main(String[] args) {

        int testScore;
        char grade;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        testScore = inputValue.nextInt();

        if (testScore >= 90) {
            grade = 'A';
        } else {
            if (testScore >= 80) {
                grade = 'B';
            } else {
                if (testScore >= 70) {
                    grade = 'C';
                } else {
                    if (testScore >= 60) {
                        grade = 'D';
                    } else {
                        grade = 'F';
                    }
                }
            }
        }

        System.out.println("Grade = " + grade);
    }
}
```

if-else

```
import java.util.Scanner;
class IfElseDemo {
    public static void main(String[] args) {

        int testScore;
        char grade;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        testScore = inputValue.nextInt();

        if (testScore >= 90) {
            grade = 'A';
        } else {
            if (testScore >= 80) {
                grade = 'B';
            } else {
                if (testScore >= 70) {
                    grade = 'C';
                } else {
                    if (testScore >= 60) {
                        grade = 'D';
                    } else {
                        grade = 'F';
                    }
                }
            }
        }

        System.out.println("Grade = " + grade);
    }
}
```

if-else

```
import java.util.Scanner;
class IfElseDemo {
    public static void main(String[] args) {

        int testScore;
        char grade;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        testScore = inputValue.nextInt();

        if (testScore >= 90) {
            grade = 'A';
        } else {
            if (testScore >= 80) {
                grade = 'B';
            } else {
                if (testScore >= 70) {
                    grade = 'C';
                } else {
                    if (testScore >= 60) {
                        grade = 'D';
                    } else {
                        grade = 'F';
                    }
                }
            }
        }

        System.out.println("Grade = " + grade);
    }
}
```

if-else

```
import java.util.Scanner;
class IfElseDemo {
    public static void main(String[] args) {

        int testScore;
        char grade;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        testScore = inputValue.nextInt();

        if (testScore >= 90) {
            grade = 'A';
        } else if (testScore >= 80) {
            grade = 'B';
        } else if (testScore >= 70) {
            grade = 'C';
        } else if (testScore >= 60) {
            grade = 'D';
        } else {
            grade = 'F';
        }

        System.out.println("Grade = " + grade);
    }
}
```

Switch

```
import java.util.Scanner;
class SwitchDemo {
    public static void main(String[] args) {
        int month;
        //Reading the value
        Scanner inputValue;
        inputValue = new Scanner(System.in);
        month = inputValue.nextInt();

        String monthString;
        switch (month) {
            case 1: monthString = "January";
                    break;
            case 2: monthString = "February";
                    break;
            ...
            case 11: monthString = "November";
                    break;
            case 12: monthString = "December";
                    break;
            default: monthString = "Invalid month";
                    break;
        }
        System.out.println(monthString);
    }
}
```

Switch

```
import java.util.Scanner;
public class SwitchDemo {
    public static void main(String[] args) {
        int month;
        //Reading the value
        Scanner inputValue;
        inputValue = new Scanner(System.in);
        month = inputValue.nextInt();

        String monthString;
        switch (month) {
            case 1: monthString = "January";
                    break;
            case 2: monthString = "February";
                    break;
            ...
            case 11: monthString = "November";
                     break;
            case 12: monthString = "December";
                     break;
            default: monthString = "Invalid month";
                     break;
        }
        System.out.println(monthString);
    }
}
```

Switch

Ejercicio 3

Crea un nuevo
Proyecto
SwitchDemo y
compártelo en
Github

```
import java.util.Scanner;
public class SwitchDemo {
    public static void main(String[] args) {
        int month;
        //Reading the value
        Scanner inputValue;
        inputValue = new Scanner(System.in);
        month = inputValue.nextInt();

        String monthString;
        switch (month) {
            case 1: monthString = "January";
                    break;
            case 2: monthString = "February";
                    break;
            ...
            case 11: monthString = "November";
                    break;
            case 12: monthString = "December";
                    break;
            default: monthString = "Invalid month";
                    break;
        }
        System.out.println(monthString);
    }
}
```

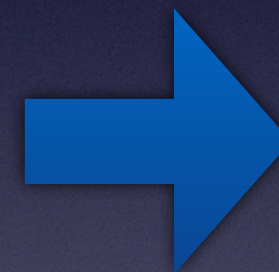
Switch

```
public class SwitchDemoFallThrough {  
    public static void main(String[] args) {  
  
        int month = 8;  
  
        switch (month) {  
            case 1: System.out.println("January");  
            case 2: System.out.println("February");  
            case 3: System.out.println("March");  
            case 4: System.out.println("April");  
            case 5: System.out.println("May");  
            case 6: System.out.println("June");  
            case 7: System.out.println("July");  
            case 8: System.out.println("August");  
            case 9: System.out.println("September");  
            case 10: System.out.println("October");  
            case 11: System.out.println("November");  
            case 12: System.out.println("December");  
                    break;  
            default: break;  
        }  
    }  
}
```

Switch

```
public class SwitchDemoFallThrough {  
    public static void main(String[] args) {  
  
        int month = 8;  
  
        switch (month) {  
            case 1: System.out.println("January");  
            case 2: System.out.println("February");  
            case 3: System.out.println("March");  
            case 4: System.out.println("April");  
            case 5: System.out.println("May");  
            case 6: System.out.println("June");  
            case 7: System.out.println("July");  
            case 8: System.out.println("August");  
            case 9: System.out.println("September");  
            case 10: System.out.println("October");  
            case 11: System.out.println("November");  
            case 12: System.out.println("December");  
                    break;  
            default: break;  
        }  
    }  
}
```

Si olvidas los
break



August
September
October
November
December

Pruébalo

Switch

```
import java.util.Scanner;
class SwitchDemo2 {
    public static void main(String[] args) {
        int year = 2000;
        int numDays = 0;
        int month;
        //Reading the value
        Scanner inputValue;
        inputValue=new Scanner(System.in);
        month = inputValue.nextInt();
        switch (month) {
            case 1: case 3: case 5:
            case 7: case 8: case 10:
            case 12:
                numDays = 31;
                break;
            case 4: case 6:
            case 9: case 11:
                numDays = 30;
                break;
            case 2:
                if (((year % 4 == 0) &&
                    !(year % 100 == 0))
                    || (year % 400 == 0))
                    numDays = 29;
                else
                    numDays = 28;
                break;
            default:
                System.out.println("Invalid month.");
                break;
        }
        System.out.println("Number of Days = "
                           + numDays);
    }
}
```

While

```
while (expression) {  
    statements  
}
```

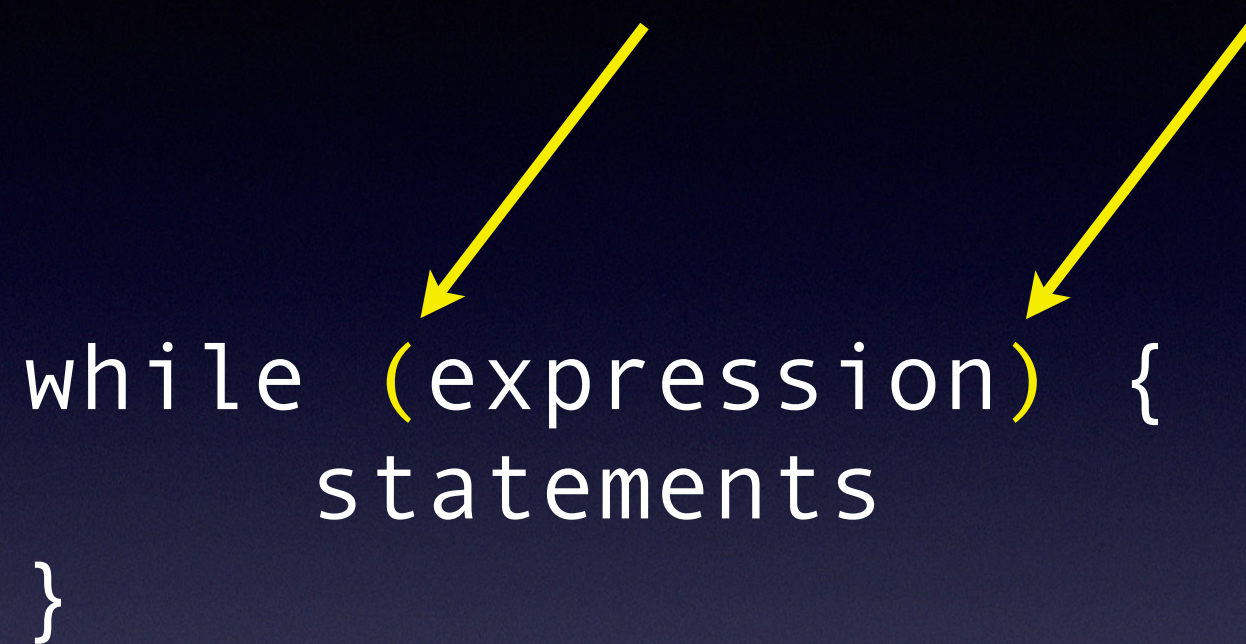
While



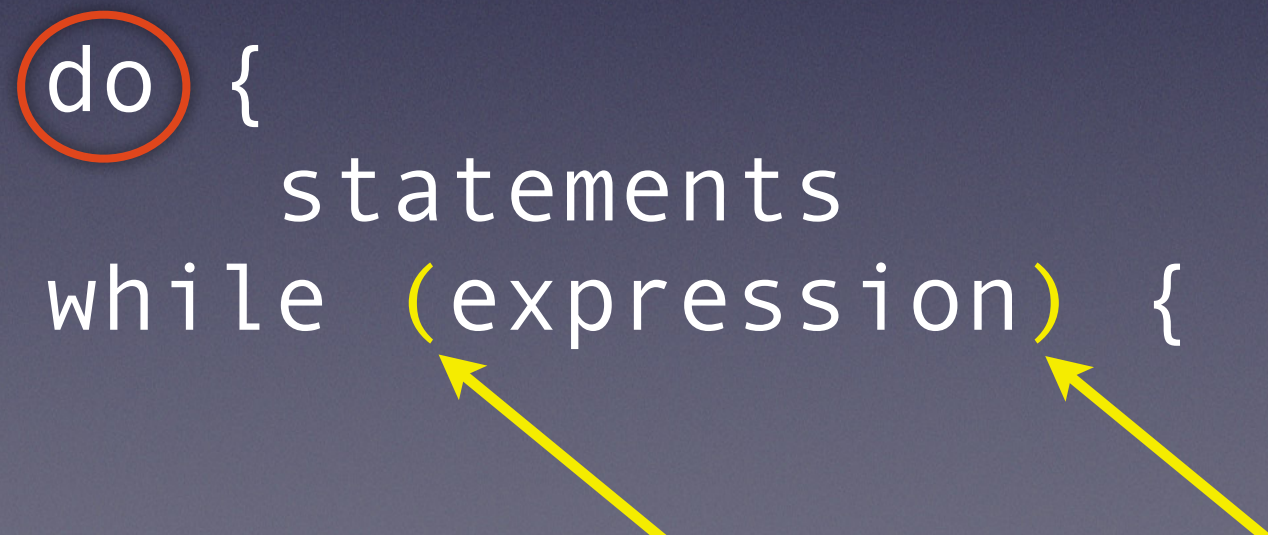
```
while (expression) {  
    statements  
}
```

The diagram illustrates the mapping from the word "While" to its syntactic components in a while loop. Two yellow arrows originate from the word "While" at the top. The left arrow points to the keyword "while" in the code snippet below. The right arrow points to the "expression" part of the "(expression)" in the same code snippet. The code snippet is formatted with "while" and "statements" on one line, "(expression)" and "{" on the next line, and "}" on the third line.

While



```
while (expression) {  
    statements  
}
```



```
do {  
    statements  
    while (expression) {
```

While

```
class WhileDemo {  
    public static void main(String[] args){  
        int count = 1;  
        while (count < 11) {  
            System.out.println("Count is: " + count);  
            count++;  
        }  
    }  
}
```

While

```
class WhileDemo {  
    public static void main(String[] args){  
        int count = 1;  
        while (count < 11) {  
            System.out.println("Count is: " + count);  
            count++;  
        }  
    }  
}
```

Do-While

```
class DoWhileDemo {  
    public static void main(String[] args){  
        int count = 1;  
        do {  
            System.out.println("Count is: " + count);  
            count++;  
        } while (count < 11);  
    }  
}
```

Do-While

```
class DoWhileDemo {  
    public static void main(String[] args){  
        int count = 1;  
        do {  
            System.out.println("Count is: " + count);  
            count++;  
        } while (count < 11);  
    }  
}
```

For

```
for (initialization; condition; increment) {  
    statements  
}
```

For

```
for (initialization; condition; increment) {  
    statements  
}
```

```
class ForDemo {  
    public static void main(String[] args){  
        for(int i = 1; i < 11; i++) {  
            System.out.println("Count is: " + i);  
        }  
    }  
}
```

For

```
for (initialization; condition; increment) {  
    statements  
}
```

```
class ForDemo {  
    public static void main(String[] args){  
        for(int i = 1; i < 11; i++) {  
            System.out.println("Count is: " + i);  
        }  
    }  
}
```

break

```
import java.util.Scanner;
class BreakDemo {
    public static void main(String[] args) {
        boolean prime = true;
        int n;
        //Reading the value
        Scanner inputValue;
        inputValue = new Scanner(System.in);
        n = inputValue.nextInt();

        for (int i = 2; i < n; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }
        if (prime) {
            System.out.println("Prime");
        } else {
            System.out.println("NOT Prime");
        }
    }
}
```

break

```
import java.util.Scanner;
class BreakDemo {
    public static void main(String[] args) {
        boolean prime = true;
        int n;
        //Reading the value
        Scanner inputValue;
        inputValue = new Scanner(System.in);
        n = inputValue.nextInt();

        for (int i = 2; i < n; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }
        if (prime) {
            System.out.println("Prime");
        } else {
            System.out.println("NOT Prime");
        }
    }
}
```



Sale del bucle más interno

continue

```
class ContinueDemo {  
    public static void main(String[] args) {  
  
        String searchMe = "peter piper picked a " +  
                           "peck of pickled peppers";  
        int max = searchMe.length();  
        int numPs = 0;  
  
        for (int i = 0; i < max; i++) {  
            // interested only in p's  
            if (searchMe.charAt(i) != 'p')  
                continue;  
  
            // process p's  
            numPs++;  
        }  
        System.out.println("Found " + numPs +  
                           " p's in the string.");  
    }  
}
```

Va a la siguiente
iteración del
bucle



return

```
class MyClass {  
    ...  
  
    void changeCadence() {  
        sentence 1;  
        sentence 2;  
        return;  
        sentence 3;  
    }  
  
    int changeGear() {  
        int i=0;  
        ...  
        sentence 1;  
        sentence 2;  
        return i+8;  
        sentence 3;  
    }  
}
```

Sale del método

Arrays

Primer índice

Índice 8

Elemento en el índice 8

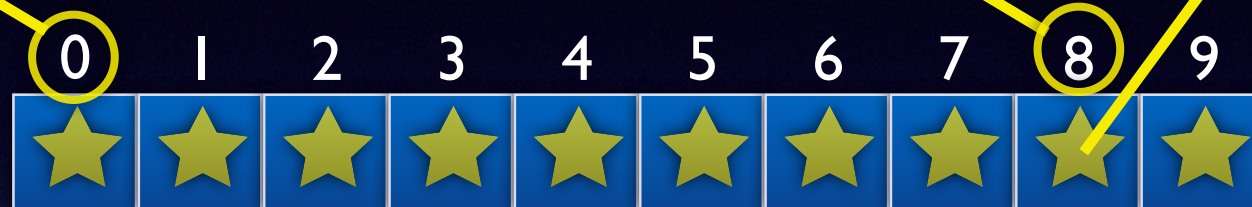


Arrays

Primer índice

Índice 8

Elemento en el índice 8



← Longitud del array es 10 →



Arrays

Primer índice

Índice 8

Elemento en el índice 8



Declaración

```
int[ ] myArray;
```



Arrays

Primer índice

Índice 8

Elemento en el índice 8



Declaración

`int[ ] myArray;` ← Forma recomendada

`int myArray[ ];`

Arrays



Declaración

```
int[ ] myArray;
```

```
int myArray[ ];
```

Instanciación

```
myArray = new int[10];
```

```
class ArrayDemo {  
    public static void main(String[] args) {  
        // declares an array of integers  
        int[] anArray;  
  
        // allocates memory for 10 integers  
        anArray = new int[10];
```

```
        // initialize first element  
        anArray[0] = 100;  
        // initialize second element  
        anArray[1] = 200;  
        // and so forth  
        anArray[2] = 300;  
        anArray[3] = 400;  
        anArray[4] = 500;  
        anArray[5] = 600;  
        anArray[6] = 700;  
        anArray[7] = 800;  
        anArray[8] = 900;  
        anArray[9] = 1000;
```

```
        System.out.println("Element at index 0: "  
                            + anArray[0]);  
        System.out.println("Element at index 1: "  
                            + anArray[1]);  
        System.out.println("Element at index 2: "  
                            + anArray[2]);  
        System.out.println("Element at index 3: "  
                            + anArray[3]);  
        System.out.println("Element at index 4: "  
                            + anArray[4]);
```

Arrays



→ **Inicialización**

```

class ArrayDemo {
    public static void main(String[] args) {
        // declares an array of integers
        int[] anArray;

        // allocates memory for 10 integers
        anArray = new int[10];

        // initialize first element
        anArray[0] = 100;
        // initialize second element
        anArray[1] = 200;
        // and so forth
        anArray[2] = 300;
        anArray[3] = 400;
        anArray[4] = 500;
        anArray[5] = 600;
        anArray[6] = 700;
        anArray[7] = 800;
        anArray[8] = 900;
        anArray[9] = 1000;

        System.out.println("Element at index 0: "
            + anArray[0]);
        System.out.println("Element at index 1: "
            + anArray[1]);
        System.out.println("Element at index 2: "
            + anArray[2]);
        System.out.println("Element at index 3: "
            + anArray[3]);
        System.out.println("Element at index 4: "
            + anArray[4]);
    }
}

```

Arrays



Salida

```

Element at index 0: 100
Element at index 1: 200
Element at index 2: 300
Element at index 3: 400
Element at index 4: 500
Element at index 5: 600
Element at index 6: 700
Element at index 7: 800
Element at index 8: 900
Element at index 9: 1000

```

Arrays



```
class ArrayDemo2 {  
    public static void main(String[] args) {  
        // declares an array of integers  
        int[] anArray;  
  
        // allocates memory for 10 integers  
        anArray = new int[10];  
  
        for (int i = 0; i < 10; i++)  
            anArray[i] = (i + 1) * 100;  
  
        for (int i = 0; i < 10; i++)  
            System.out.println("Element at index " + i + ": "  
                               + anArray[i]);  
    }  
}
```

Salida

```
Element at index 0: 100  
Element at index 1: 200  
Element at index 2: 300  
Element at index 3: 400  
Element at index 4: 500  
Element at index 5: 600  
Element at index 6: 700  
Element at index 7: 800  
Element at index 8: 900  
Element at index 9: 1000
```

Arrays

Primer índice

Índice 8

Elemento en el índice 8



```
class ArrayDemo3 {  
    public static void main(String[] args) {  
        // declares an array of integers and initializes  
        int[] anArray = {101, 200, 305, 400, 500, 600,  
                        700, 800, 908, 1000 };  
        for (int i = 0; i < 10; i++)  
            System.out.println("Element at index " + i + ": "  
                               + anArray[i]);  
    }  
}
```

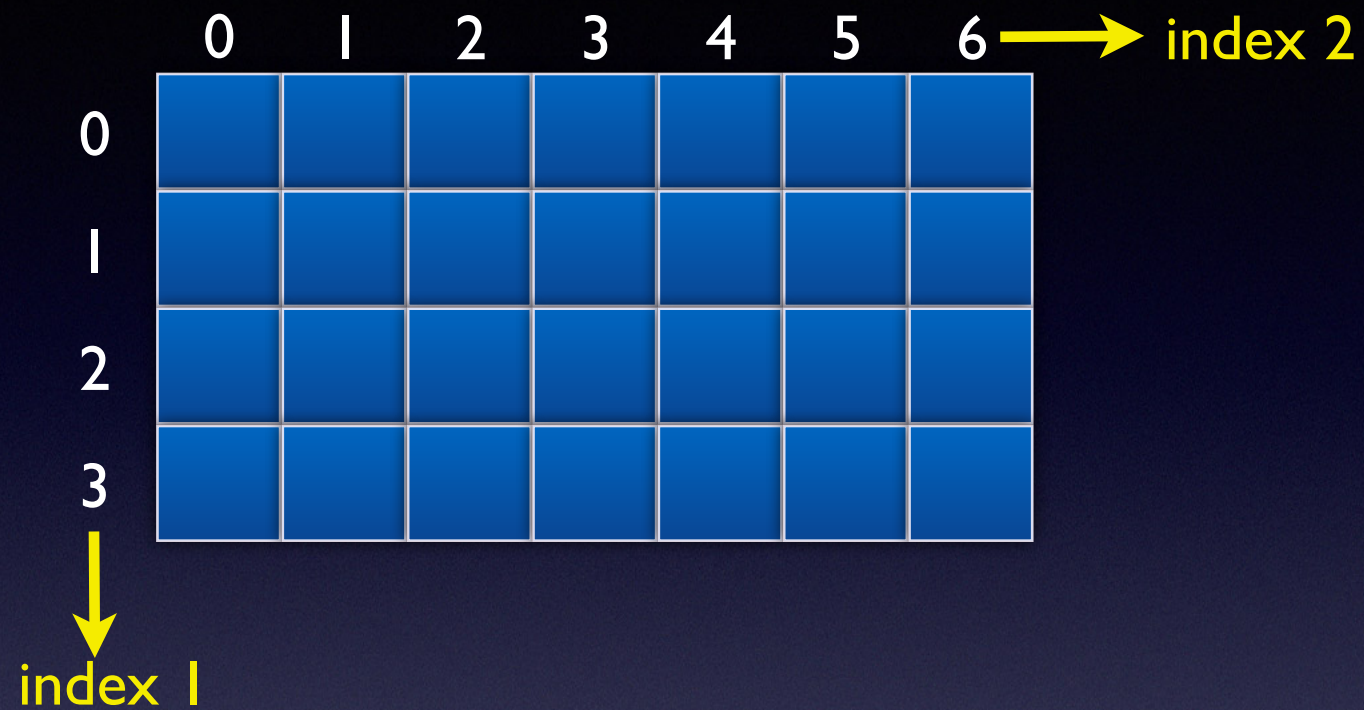
Salida

```
Element at index 0: 101  
Element at index 1: 200  
Element at index 2: 305  
Element at index 3: 400  
Element at index 4: 500  
Element at index 5: 600  
Element at index 6: 700  
Element at index 7: 800  
Element at index 8: 908  
Element at index 9: 1000
```

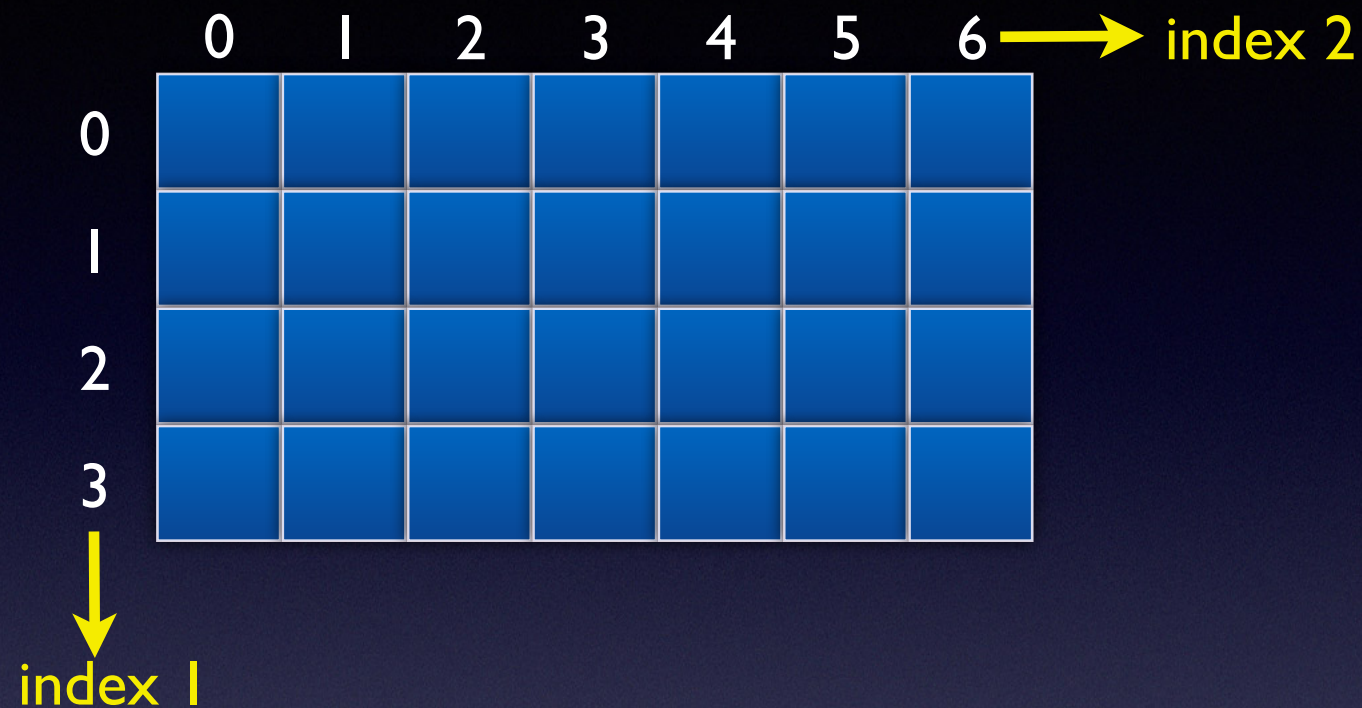
Arrays Multidimensionales

	0	1	2	3	4	5	6
0	★	★	★	★	★	★	★
1	★	★	★	★	★	★	★
2	★	★	★	★	★	★	★
3	★	★	★	★	★	★	★

Arrays Multidimensionales



Arrays Multidimensionales



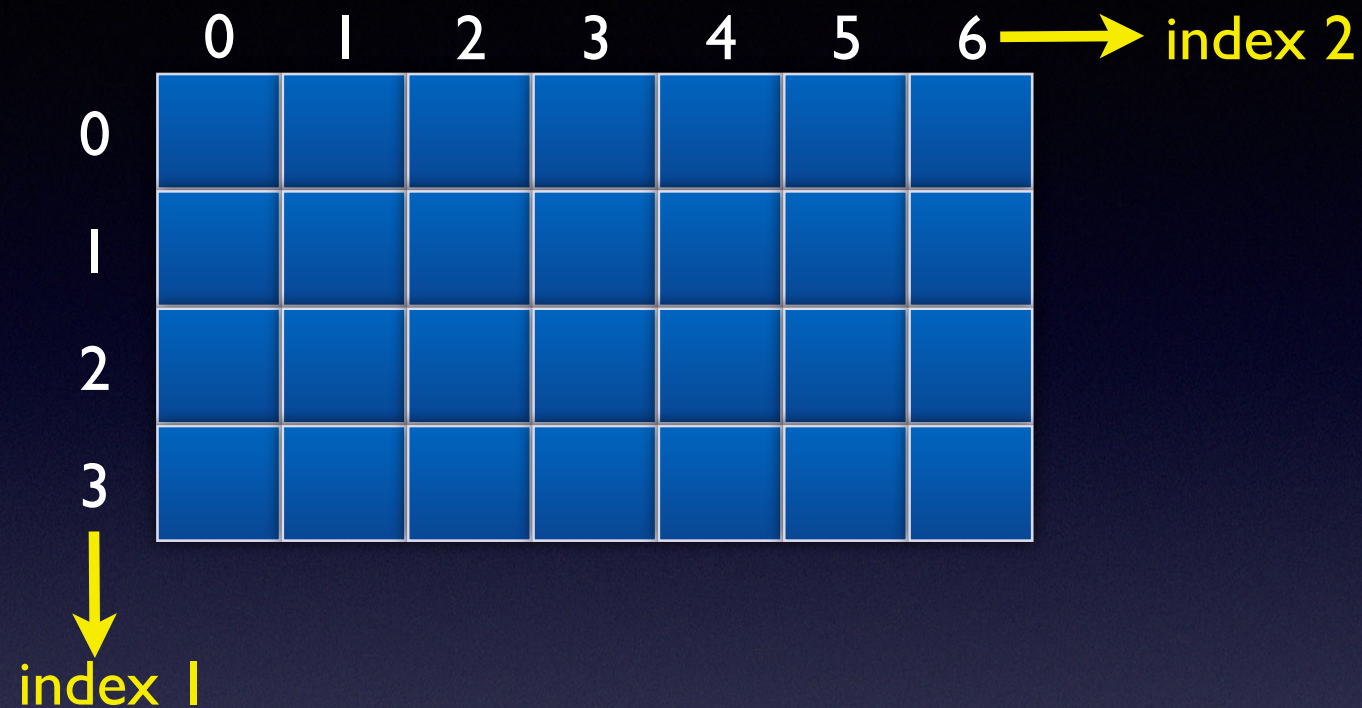
Declaración

```
int[ ][ ] myMatrix;
```

Instanciación

```
myMatrix = new int[4][7];
```

Arrays Multidimensionales



Declaración

```
int[ ][ ] myMatrix;
```

Instanciación

```
myMatrix = new int[4][7];
```

Arrays Multidimensionales

	0	1	2	3	4	5	6	→ index 2
0	1	2	3	4	5	6	7	
1	8	9	10	11	12	13	14	
2	15	16	17	18	19	20	21	
3	22	23	24	25	26	27	28	

↓ index 1

Arrays Multidimensionales

	0	1	2	3	4	5	6
0	1	2	3	4	5	6	7
1	8	9	10	11	12	13	14
2	15	16	17	18	19	20	21
3	22	23	24	25	26	27	28

 **i**

 **index j**

```
class ArrayDemo4 {  
    public static void main(String[] args) {  
  
        int[][] anArray = new int[4][7];  
        int cont = 1;  
  
        for (int i = 0; i < 4; i++) {  
            for (int j = 0; j < 7; j++) {  
                anArray[i][j] = cont;  
                cont++;  
            }  
        }  
        for (int i = 0; i < 4; i++) {  
            for (int j = 0; j < 7; j++) {  
                System.out.println("Element at index " +  
                                    i + "," + j + ": " +  
                                    anArray[i][j]);  
            }  
        }  
    }  
}
```

Arrays Multidimensionales

	0	1	2	3	4	5	6
0	1	2	3	4	5	6	7
1	8	9	10	11	12	13	14
2	15	16	17	18	19	20	21
3	22	23	24	25	26	27	28

↓ *i*

→ index *j*

```
class ArrayDemo4 {
    public static void main(String[] args) {

        int[][] anArray = new int[4][7];
        int cont = 1;

        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 7; j++) {
                anArray[i][j] = cont;
                cont++;
            }
        }

        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 7; j++) {
                System.out.println("Element at index " +
                                    i + "," + j + ": " +
                                    anArray[i][j]);
            }
        }
    }
}
```

Ejercicio 4

Sube ArrayDemo4

Arrays Multidimensionales

	0	1	2	3	4	5	6
0	1	2	3	4	5	6	7
1	8	9	10	11	12	13	14
2	15	16	17	18	19	20	21
3	22	23	24	25	26	27	28

↓
i

→ index *j*

```
class ArrayDemo5 {  
    public static void main(String[] args) {  
  
        int[][] anArray = { {1,2,3,-4,5,6,7},  
                             {8,9,10,-11,12,13,14},  
                             {15,16,17,-18,19,20,21},  
                             {22,23,24,-25,26,27,28} };  
  
        for (int i = 0; i < 4; i++) {  
            for (int j = 0; j < 7; j++) {  
                System.out.println("Element at index "  
                                    + i + "," + j + ": " +  
                                    anArray[i][j]);  
            }  
        }  
    }  
}
```

Copiando Arrays

0	1	2	3	4	5	6	7	8	9	10	11	12
d	e	c	a	f	f	e	i	n	a	t	e	d

```
class ArrayCopyDemo {  
    public static void main(String[] args) {  
        char[] copyFrom = { 'd', 'e', 'c', 'a', 'f', 'f',  
                             'e', 'i', 'n', 'a', 't', 'e', 'd' };  
        char[] copyTo = new char[7];  
  
        System.arraycopy(copyFrom, 2, copyTo, 0, 7);  
        System.out.println(new String(copyTo));  
    }  
}
```

Copiando Arrays

0	1	2	3	4	5	6	7	8	9	10	11	12
d	e	c	a	f	f	e	i	n	a	t	e	d

```
class ArrayCopyDemo {  
    public static void main(String[] args) {  
        char[] copyFrom = { 'd', 'e', 'c', 'a', 'f', 'f',  
                             'e', 'i', 'n', 'a', 't', 'e', 'd' };  
        char[] copyTo = new char[7];  
  
        System.arraycopy(copyFrom, 2, copyTo, 0, 7);  
        System.out.println(new String(copyTo));  
    }  
}
```

0	1	2	3	4	5	6

Copiando Arrays

0	1	2	3	4	5	6	7	8	9	10	11	12
d	e	c	a	f	f	e	i	n	a	t	e	d

```
class ArrayCopyDemo {  
    public static void main(String[] args) {  
        char[] copyFrom = { 'd', 'e', 'c', 'a', 'f', 'f',  
                             'e', 'i', 'n', 'a', 't', 'e', 'd' };  
        char[] copyTo = new char[7];  
  
        System.arraycopy(copyFrom, 2, copyTo, 0, 7);  
        System.out.println(new String(copyTo));  
    }  
}
```

0	1	2	3	4	5	6
c	a	f	f	e	i	n

Copiando Arrays

0	1	2	3	4	5	6	7	8	9	10	11	12
d	e	c	a	f	f	e	i	n	a	t	e	d

```
class ArrayCopyDemo {  
    public static void main(String[] args) {  
        char[] copyFrom = { 'd', 'e', 'c', 'a', 'f', 'f',  
                             'e', 'i', 'n', 'a', 't', 'e', 'd' };  
        char[] copyTo = new char[7];  
  
        System.arraycopy(copyFrom, 2, copyTo, 0, 7);  
        System.out.println(new String(copyTo));  
    }  
}
```

0	1	2	3	4	5	6
c	a	f	f	e	i	n

7 elementos

Copiando Arrays

0	1	2	3	4	5	6	7	8	9	10	11	12
d	e	c	a	f	f	e	i	n	a	t	e	d

```
import java.util.Arrays;
class ArrayCopyOfDemo {
    public static void main(String[] args) {

        char[] copyFrom = {'d', 'e', 'c', 'a', 'f', 'f', 'e',
                           'i', 'n', 'a', 't', 'e', 'd'};
        char[] copyTo = Arrays.copyOfRange(copyFrom, 2, 9);
        System.out.println(new String(copyTo));
    }
}
```

0	1	2	3	4	5	6
c	a	f	f	e	i	n

Cuidado!!!
Índice final excluido -> 8

Manipulando Arrays

Otros métodos interesante en la clase `java.util.Arrays`

- `binarySearch()`
- `equals()`
- `fill()`
- `sort()`

Ejercicios

- Desarrolla un algoritmo que lea 10 números introducidos desde el teclado. Guárdalos en un array. Luego, muéstralos en orden inverso.

Ejercicio 5

Sube Exercise5

Ejercicios

- Desarrolla un algoritmo que lea 10 números introducidos desde el teclado. Guárdalos en un array. Luego, invierte el array e imprímelo en orden normal.

Nota: Para invertir la matriz, intercambie los elementos: el elemento en la posición 0 con el elemento en la posición 9, el elemento en la posición 1 con el elemento en la posición 8, y así sucesivamente.

Ejercicio 6

Sube Exercise6

Ejercicios

Escribe un programa que lea un DNI (sin la letra) y luego calcule la letra, sabiendo que la letra se calcula de esta manera:

Toma el DNI como un número y calcula el resto de la división entre 23, y asigna la letra en función del resto usando la siguiente tabla.:

Resto:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
T	R	W	A	G	M	Y	F	P	D	X	B	N	J	Z	S	Q	V	H	L	C	K	E

Ejercicio 7

Sube Exercise7

Ejercicios

Escribe un programa que cree una matriz de 10 x 10 elementos. Inicialice sus elementos a 1 y luego establezca a 8 los elementos (0, 4), (2, 6), (3, 1) y (8, 6). Y luego la imprima así:

				8					
						8			
	8								
						8			

Ejercicio 8

Sube Exercise8

Ejercicios

Modifica el programa anterior para mostrar cuántas filas hay con todos los elementos a 1 y cuántas columnas hay con todos los elementos a 1.

				8					
						8			
	8								
						8			

Ejercicio 9

Sube Exercise9

Salida: Rows | = 6 / Columns | = 7

Ejercicios

- Escribe un programa que calcule la moda (el elemento más frecuente) de una lista de 20 caracteres. Inicializa el array de caracteres en la declaración del array como una lista de caracteres como esta.:

```
final char[] LETTERS =  
    {'B', 'A', 'B', 'C', 'A', 'C', 'D', 'C', ..., 'C', 'D'};
```

Ejercicio 10

Sube Exercise10

1. Escribe un programa que simule el juego "Batalla Naval" sobre un tablero de 8x8. La computadora colocará 10 barcos al azar (de una sola celda). El usuario introducirá una coordenada (primera letra y luego un número) y el programa escribirá el tablero completo cada vez, con los barcos hundidos (X) y los disparos fallidos (círculos). Mostrará el contador de disparos y el número de barcos hundidos.

Ejemplo:

SHOTS: 12

SUNK SHIPS: 3

	1	2	3	4	5	6	7	8
A								
B		O		O		O		
C		O					O	
D	X	O						
E				X	O		O	
F	O			O				
G	O				O		X	
H								O

Enter row (Letter): C

Enter column (Number): 2

Ejercicio I I

Sube Exercise I I

Accediendo a elementos de un array con un bucle For

```
class EnhancedForDemo {  
    public static void main(String[] args){  
        int[] numbers =  
            {1,2,3,4,5,6,7,8,9,10};  
        for (int item : numbers) {  
            System.out.println("Count is: " + item);  
        }  
    }  
}
```

Lee esto como: **for int item in numbers**

Ejercicio Select-Sort

Algoritmo Select-Sort para ordenar un array

En este método, buscamos el elemento más pequeño y lo insertamos en la primera posición del array. Luego, con el segundo elemento, realizamos el mismo procedimiento y repetimos el mismo procedimiento para los demás elementos.

Puedes ver una representación del algoritmo en este video:
<https://www.youtube.com/watch?v=Ns4TPTC8whw>

Ejercicio

Implementa el algoritmo Select-Sort en Java. Pruébalo con un array de cadenas inicializadas en el programa (sin leerlas con el teclado).

Ejercicio Bubble-Sort

Ordenamiento de Burbuja

En este método, se comparan el primer y el segundo elemento. Si el primero es mayor que el segundo, intercambian posiciones. Luego, se comparan el segundo y el tercero, y se continúa comparando hasta llegar al final del array. El último elemento se ordenará. En la siguiente iteración, se sigue el mismo procedimiento, pero se llega solo al elemento $n-2$. En la siguiente iteración, se continúa hasta el elemento $n-3$, hasta llegar al elemento 0.

Este procedimiento suele obtener un array ordenado antes de llegar a la iteración del elemento 1, por lo que se puede mejorar detectando si no se produjo ningún intercambio. En ese caso, el array ya está ordenado.

Puedes ver una representación del algoritmo en este video:

[http://www.youtube.com/watch?](http://www.youtube.com/watch?v=lyZQPjUT5B4&list=PLP2eLJiSCJbAclmdE6sqx30XphnZZ4Rjo)

[v=lyZQPjUT5B4&list=PLP2eLJiSCJbAclmdE6sqx30XphnZZ4Rjo](http://www.youtube.com/watch?v=lyZQPjUT5B4&list=PLP2eLJiSCJbAclmdE6sqx30XphnZZ4Rjo)

Ejercicio

Implementa el algoritmo Ordenamiento de Burbuja en Java. Pruébalo con un array de Strings inicializado en el programa (sin leerlos mediante el teclado).